Zero-parameter structural computation of Standard Model observables and constants and a dark-matter tower from $E_8 \times E_8$ Coxeter geometry on a Schoen Calabi–Yau threefold

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What this paper does. We construct the Standard Model fermion mass spectrum, gauge couplings, mixing angles (CKM, PMNS), CP phases, dark-matter mass tower, Higgs boson mass, Higgs vacuum expectation value, and Newton's gravitational constant from a small set of fundamental equations acting on the fixed geometry of $E_8 \times E_8$ heterotic compactification on the Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$ Calabi–Yau threefold quotient $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$, with the McKay correspondence on the binary icosahedral group $2I$ and Wilson $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry breaking. The construction has no PDG inputs and no adjustable quantities.

Why it matters. Approximately twenty-six SM observables are predicted to ppm-or-better precision: m_e at 17 ppm, m_μ at 30 ppb, m_τ at 0.15 ppm, $\sin^2\theta_W = 2/9 + \Delta_\varphi/h$ at 198 ppm, $\alpha_s(M_Z) = 849/7192$ at 405 ppm, the dark-matter abundance $\Omega_{\rm DM}h^2 = 0.119982$ vs. Planck 2018 0.1200(10), and Newton's G to 2.5 ppm vs. CODATA 2022 — all from the geometry alone. The CP phases $\delta_{\rm CP} = \pi + \pi/10$ and $\delta_{\rm CKM} = 11\pi/30$ are predicted as rational multiples of π. Normal neutrino mass ordering is selected. A conditional geometric route to vanishing strong CP via the $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient projection avoids invoking a Peccei–Quinn axion.

What is open and not claimed. The construction is a strong-selection structural identification, not yet a forced derivation: the substrate-physical projection onto the heterotic moduli of $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ is not closed in this paper. The σ-discrimination Schoen-vs-Tian–Yau result (6.92σ Tier 0; bundle-construction-specific) is reported as supporting evidence pending the converged PDG-fermion-mass Yukawa-cohomology pipeline. Dark-sector chirality assignments and several closure-rule numerators are encoded structural-rule-consistent choices whose generative derivation is deferred.

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I. INTRODUCTION

The Standard Model (SM) of particle physics, despite its empirical success, contains twenty-six parameters whose values are not predicted by any underlying principle but determined empirically from observation [1]. These include the three gauge couplings, the Higgs sector parameters (vacuum expectation value and self-coupling), nine charged-fermion Yukawa couplings, three neutrino mass scales, four quark-sector mixing parameters (CKM), four lepton-sector mixing parameters (PMNS), and the QCD vacuum angle $\theta_{\rm QCD}$. Beyond the SM, the existence and mass scale of dark matter constitutes one of the most prominent unresolved puzzles in contemporary physics [2].

This paper presents a structural computation, with no fitted continuous parameter, of all twenty-six SM parameters, Newton's gravitational constant, the Planck mass, and a discrete dark-matter mass tower with associated decay constant, couplings, and cosmological abundance, given a fixed set of first-principles starting assumptions (Table I). Per-particle Coxeter chain indices, sign assignments, sector and cycle placements, and rational closure rules arise as structural outputs of those starting commitments rather than as independently varied parameters; the audit is in Sec. I 0 b. PDG-matching residuals are predictions versus measurement, not fits. The framework's predictive path takes *zero* measured masses as input. The Higgs vacuum expectation value is computed in closed form from the E_8 Coxeter invariants alone via the structural identity $v_C = (\dim -2)/(1-\delta_W)\cdot(1+5\,\alpha_\tau^3/8)$ (see substrate_mass_spectrum.py:402; inputs $\dim = 248$ at line 160, $h = 30$ at line 161, $m_{\max} = 29$ at line 163, $e_6 = 19$ at line 399, and the derived $\delta_W = 1/((h+m_{\max})e_6) = 1/1121$ at line 400 and $\alpha_\tau = 1/(\dim + h) = 1/278$ at line 188), giving $v_C = 246.21965$ GeV. The gateway tension scale is then $h_{\rm eff} = v_C/R_v$ where R_v is the structural Higgs-vev ratio, and every other mass is the corresponding chain-specific structural ratio times $h_{\rm eff}$ (*φ-chain leptons, $\sqrt{2}$-chain quarks/EW bosons, α_τ-tower neutrinos and dark-matter modes*). As an internal-arithmetic self-consistency check, alternatively anchoring the framework on the CODATA electron mass alone or muon mass alone produces predictions that agree at 0.0298 ppm ($\sim$ 30 ppb), set by sub-harmonic-series truncation; both fermion-anchored modes agree with the structural-vev anchor to within 0.12 ppm on every observable. This two-anchor *diagnostic* validates the framework's internal arithmetic but is not the predictive path.

a. Starting assumptions of this paper. Table I enumerates the complete list of first-principles commitments on which the numerical demonstration of this paper rests. The list contains no continuous fitted parameter. Every structural integer, sign, sub-harmonic factor, and closure rule that appears in the mass equations of Secs. III–XI is either an immediate consequence of these commitments or, where so flagged, an independent structural rule whose derivation is the subject of the companion paper [43].

b. Structural-output audit. The construction generates discrete data — per-particle Coxeter chain indices, sign assignments, sector and cycle placements, and a family of closure rules — as *structural outputs* of the starting commitments of Table I, not as independently varied parameters. They are not free: changing any one of them in isolation would generally be inconsistent with the underlying structural rules (the binary-icosahedral McKay seed, the φ- and $\sqrt{2}$-chain harmonic-recursion split, the boundary capacity $h + m_{\max} = 59$, the E_8 Coxeter spectrum, and the Wilson-line-broken visible chain). They become arbitrary only under the counterfactual in which the structure is removed. We list the discrete data explicitly so the reader can audit by inspection which items are derived in closed form within this paper and which are encoded as structural-rule-consistent choices whose generative derivation is part of the open program.

- *Per-particle Coxeter chain index k_X* for each charged fermion, EW boson, and the Higgs vev

TABLE I. Starting assumptions of the present paper. The numerical demonstration in Secs. III–XI takes these as given. The argument that this list is minimal — i.e. that one would expect the substrate-physical principles to select these and not others — is the subject of the companion paper [43] and is not load-bearing for the present demonstration.

Starting assumption	Status / role in the present paper
Heterotic $E_8 \times E_8$ string theory [11]	Standard heterotic literature; assumed.
Schoen Calabi–Yau threefold quotient $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ [10, 12, 13]	Empirically identified within the framework's narrowed candidate set; see Sec. II B.
E_8 Coxeter spectrum (rank 8, $h = 30$, exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$, $\lvert\Phi\rvert = 240$)	Standard Lie-algebraic data [6]; assumed.
McKay correspondence [5]; binary icosahedral McKay-dual of E_8	Standard; the icosahedral order-5 assignment as the framework's harmonic seed is a structural commitment. Companion paper argues for its substrate-physical inevitability.
Wilson-line breaking $E_8 \to G_{\rm SM}$ along $\pi_1(X/\Gamma) = \mathbb{Z}_3 \times \mathbb{Z}_3$; phenomenological lepton/quark sector split via φ-chain (5-fold icosahedral) and $\sqrt{2}$-chain (4-fold octahedral) harmonic recursions	Standard heterotic chain $E_8 \supset E_6 \supset SO(10) \supset SU(5) \supset G_{\rm SM}$. The $\varphi/\sqrt{2}$ chain split is a framework structural assignment matching observed mass hierarchies with small-integer Coxeter-step assignments; a derivation from a specific Wilson-line representation on X/Γ is part of the open program (Sec. XV).
Substrate-tension reading of the Higgs mechanism (motivated by Anderson's superconducting precursor [7]; similar in spirit to but independent of Volovik's emergent-vacuum program [8])	Interpretive frame for the Higgs vacuum; the v ladder in Sec. V is a closed-form structural identity within this reading.
Coldea *et al.* [3] confirmation of the Zamolodchikov E_8 spectrum [4] in $CoNb_2O_6$	Empirical observation of the E_8 root system in a 1+1D condensed-matter context; adopted as motivation for the framework's E_8 commitment, not as a direct measurement of the heterotic gauge E_8. Not load-bearing for the numerical computations of the present paper.

(App. A): $k_e = 0$, $k_\mu = 11$, $k_\tau = 17$ on the φ-chain (lepton positions at integer E_8 exponents); $k_u = 4$, $k_d = 6.5$, $k_s = 15$, $k_c = 22.5$, $k_b = 26$, $k_t = 36.5$, $k_W = 34.5$, $k_v = 1132/30$, $k_H = k_v - 2$ on the $\sqrt{2}$-chain. The Higgs-vev position $k_v = h + \mathrm{rank} - \mathrm{rank}/h$ is given in closed form (Eq. 21); the remaining k_X values are encoded as positions consistent with the integer/half-integer cycle-1 and cycle-2 structure of the chain. A generative derivation from the specific Wilson-line representation on X/Γ is part of the open program (Sec. XV).

- *Per-quark sign assignments* $\chi_q, \lambda_q \in \{-1, 0, +1\}$ in the quark mass formula (Table IX): encoded as a structural pattern across the six quarks; their derivation from a structural-rule generator (chirality flip across the harmonic threshold $h/2$, per-step-loss sign across cycle boundaries) is part of the open program.

- *Sector chain-type assignment* (φ vs $\sqrt{2}$): leptons and neutrinos to the φ-chain; quarks and electroweak bosons to the $\sqrt{2}$-chain. This pairing reflects the framework's McKay-icosahedral and McKay-octahedral seed assignments; an explicit derivation from a Wilson-line representation on X/Γ is open.

- *Cycle-1 vs cycle-2 placement* for each particle: encoded in the structural pattern of the chain (App. A).

- *Closure rules* $\delta_H = 1/(h + m_{\max}) = 1/59$ and $\delta_W = 1/(59 \cdot e_6) = 1/(59 \cdot 19) = 1/1121$ are given in closed form from the boundary capacity $h + m_{\max}$ and the E_8 exponent set; the gateway sub-harmonic factor $1/(4(\dim - 1))$ admits a dual-sector / two-orientation structural reading whose derivation from a heterotic-anomaly index or a $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient trace on the Schoen threefold is part of the open program.

- *Sub-harmonic correction structures* (α_v, α_H, α_W, α_t, the rank-locked sub-harmonic on m_5, and the $5\varphi^2\alpha_\tau$ correction on ν_3) are encoded as structural-rule combinations of $\{\dim, h, \mathrm{rank}, m_{\max}, e_j, \varphi\}$. A generative derivation of each from the compactification data is open.

- *Other rational closures* include $\sin^2\theta_W = 2/9 + \Delta_\varphi/h$ and $\alpha_s(M_Z) = (m_{\max}^2 + \mathrm{rank})/(m_{\max} \cdot \dim) = 849/7192$. The leading $2/9$ in $\sin^2\theta_W$ and the combination $m_{\max}^2 + \mathrm{rank}$ in α_s are framework structural choices; they are not standard E_8 Lie invariants and their derivation is open.

None of the items above is varied independently of the framework's structural rules. Items derived in closed form within this paper (boxed equations) are structurally forced; items flagged as encoded or as "framework structural assignment" are consistent with the framework's rules but their generative derivation from the compactification data is the subject of the further-exploration program (Sec. XV).

c. Epistemic posture. From a 4D observer's vantage the starting assumptions of Table I cannot be *proven*: any purported derivation of the substrate-physical content from inside the framework's macroscopic 4D regime is necessarily a consistency check, not a proof. The standard a 4D observer can apply is therefore *empirical consistency* — the assumptions are evaluated by the predictions they generate. This paper performs that evaluation by computing the SM observables and comparing to measurement. We make no claim in this paper that the starting assumptions are forced from first principles; we make the weaker, sharper claim that, taking them as given, the SM mass spectrum and mixing parameters compute directly with the residuals reported in Table I's Sec. XIII downstream. The companion paper [43] addresses the separate question of whether the starting assumptions are minimal and substrate-physically expected; that argument is independent of the present demonstration and not relied upon here.

d. Roadmap. Section II fixes the framework as commitment — heterotic structure on the Schoen quotient, E_8 Coxeter spectrum, McKay-icosahedral seed, sub-Coxeter decomposition, substrate-tension Higgs reading. Section II B reports the empirical Schoen-versus-Tian–Yau discrimination, the Tian–Yau Yukawa-channel structural exclusion, and the catalogue uniqueness sweep at Hodge $(3, 3)$, with explicit hedges for the open methodological items. Section III presents the gateway formula and the direct computation of the electron mass. Sections IV–XI present the direct computations of charged-fermion masses, the Higgs sector, electroweak gauge couplings, the strong-CP angle, neutrino masses and mixing, CKM mixing, the dark-matter Coxeter tower with cosmological abundance, and Newton's G. Section XIII presents the master results tables. Section XIV enumerates the falsifiable predictions. Section XV enumerates the open methodological items, the framework-level commitments deferred to the companion paper, and the further-exploration program. Section XVI discusses parameter-count semantics and relation to prior work.

II. THE SUBSTRATE FRAMEWORK

A. Heterotic $E_8 \times E_8$ on the Schoen quotient

The framework treats heterotic $E_8 \times E_8$ string theory [11] compactified on the Schoen Calabi–Yau threefold quotient $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ as the structural substrate. The Schoen threefold [10] is the smooth, simply-connected Calabi–Yau fiber product $X = B \times_{\mathbb{P}^1} B'$ of two rational elliptic surfaces over $\mathbb{P}^1$, with Hodge numbers $h^{1,1}(X) = h^{2,1}(X) = 19$ and Euler characteristic $\chi(X) = 0$. In the heterotic standard-model construction, suitably chosen B, B' admit a freely acting $\Gamma = \mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry, and the quotient X/Γ acquires $\pi_1(X/\Gamma) = \Gamma = \mathbb{Z}_3 \times \mathbb{Z}_3$ (while X itself remains simply-connected), enabling Wilson-line breaking of one E_8 factor to Standard-Model gauge structure [12, 13].

a. Hodge numbers and generation count. The Schoen quotient X/Γ has Hodge numbers $h^{1,1}(X/\Gamma) = h^{2,1}(X/\Gamma) = 3$, computed as the dimensions of the Γ-invariant subspaces of $H^{1,1}(X)$ and $H^{2,1}(X)$ via the character formula $\dim H^{p,q}(X)^\Gamma = (1/|\Gamma|) \sum_{g\in\Gamma} \mathrm{tr}(g^* | H^{p,q}(X))$; see [12] for the explicit representation-theoretic decomposition. The Euler characteristic of the cover, $\chi(X) = 2[h^{1,1}(X) - h^{2,1}(X)] = 0$, divides under the freely-acting Γ to give $\chi(X/\Gamma) = 0$ in absolute Euler-characteristic terms; the *net chirality index* relevant for the standard heterotic generation count is set instead by the chiral Atiyah–Singer index of the bundle V chosen to break E_8 via Wilson lines, $|n_g - \bar{n}_g| = |\mathrm{Index}(V)|/|\Gamma|$ with $\mathrm{Index}(V) = \int_X \mathrm{ch}(V) \cdot \mathrm{Td}(TX)$ (equivalently $(1/2)\int_X c_3(V)$ when $c_1(V) = 0$, by the standard Hirzebruch–Riemann–Roch reduction on a CY3). For the BHOP rank-4 $SU(4)$ extension bundle on Schoen, $\mathrm{Index}(V_{\mathrm{BHOP}}) = -27$ (BHOP-2005 Eq. 88; with $c_3/2 = -9\,\tau_1\tau_2^2$ per BHOP-2005 Eq. 97 and $\int_X \tau_1\tau_2^2 = 3$, giving Index $= -27$), and the standard Braun–He–Ovrut–Pantev model thus yields $|n_g - \bar{n}_g| = 27/9 = 3$ chiral generations as required by the visible Standard Model. The present framework assumes (rather than derives) the bundle commitment, and adopts the BHOP-2005 published index numerics.

b. Heterotic anomaly cancellation. Heterotic $E_8 \times E_8$ on a Calabi–Yau threefold requires the modified Bianchi identity $dH = (\alpha'/4)\big(\mathrm{tr}\, R \wedge R - \mathrm{tr}\, F \wedge F\big)$ to admit a globally defined H, equivalent to the cohomological condition

$$\mathrm{ch}_2(V) \;=\; \mathrm{ch}_2(TX) \;+\; [W], \tag{1}$$

where ch_2 is the second Chern character, V is the visible-sector gauge bundle, TX is the holomorphic tangent bundle of the Schoen threefold, and $[W]$ is an effective 5-brane class in $H^4(X, \mathbb{Z})$ (vanishing in the standard embedding $V = TX$). The present framework assumes that the bundle choice underlying the three-generation count is consistent with Eq. (1) for some admissible 5-brane class $[W]$, as is the case in the Braun–He–Ovrut–Pantev construction [12]. A first-principles identification of the specific $[W]$ used by the framework, together with the *slope-stability* (μ-stability) of V required by the Donaldson–Uhlenbeck–Yau theorem [27, 28] to admit a Hermitian–Yang–Mills connection on the quotient X/Γ [13], is part of the open program (Sec. XV). The Donaldson-balanced metric pipeline used in Sec. II B is constructed assuming polystability of V; the empirical convergence of the Donaldson algorithm on the candidate bundles is evidence-for, but not proof-of, polystability on the quotient.

c. Substrate dimensionality and the 10D embedding. The discrete substrate update rule developed in the companion paper [43] is specified directly in $3 + 1$ macroscopic dimensions, while the heterotic $E_8 \times E_8$ string lives in $9 + 1$. The framework's operational stance is

that the six internal dimensions are *algebraically encoded* in the substrate by the $E_8 \times E_8$ structure of the chain construction (in particular, the Coxeter-chain positions k_X and the sector-modifier $\sqrt{2}$- vs. φ-chain assignments substitute for direct propagation in the compactified directions): the substrate is treated as $3+1$D throughout, with the Calabi–Yau / heterotic structure entering only through the algebra controlling the mass spectrum and gauge structure. This is a *framework choice*, not a derivation: a substrate-physical realisation in which the $3+1$D macroscopic substrate is a boundary or low-energy slice of an underlying $9+1$D substrate (analogous in spirit to brane-world or warped-compactification constructions) is consistent with the present construction but not committed to here. In particular, no claim is made about the size, geometry, or stabilisation of the six internal directions beyond what is needed to identify the algebraic content $E_8 \times E_8$ on X/Γ.

One E_8 factor contains the visible Standard Model gauge structure after compactification. The second E_8 factor is the dark sector, communicating with the visible sector only gravitationally except through identified portal couplings.

d. Visible $E_8 \to$ SM symmetry-breaking chain. The visible E_8 is broken to the Standard-Model gauge group $G_{\rm SM} = SU(3)_{\rm color} \times SU(2)_L \times U(1)_Y$ through the BHOP $SU(4)$-structure-group chain

$$\begin{aligned} E_8 \;&\supset\; SO(10) \times SU(4) \;\supset\; SU(5) \times U(1) \\ &\supset\; G_{\rm SM} \times U(1)_X, \end{aligned} \tag{2}$$

with the final descent realised by Wilson lines along generators of $\pi_1(X/\Gamma) = \mathbb{Z}_3 \times \mathbb{Z}_3$. The bundle V committed here is the rank-4 $SU(4)$ extension construction of Braun–He–Ovrut–Pantev [14] §6 (an extension of two semi-stable summands rather than a line-bundle sum), with chiral index $\mathrm{Index}(V_{\rm BHOP}) = \int_{\widetilde{X}} \mathrm{ch}(V_{\rm BHOP}) \cdot \mathrm{Td}(T\widetilde{X}) = -27$ on the cover (BHOP-2005 Eq. 88; $c_3/2 = -9\,\tau_1\tau_2^2$ per BHOP-2005 Eq. 97), yielding three generations on the free $\Gamma = \mathbb{Z}_3 \times \mathbb{Z}_3$ quotient via the Atiyah–Singer / Hirzebruch–Riemann–Roch reduction $N_{\rm gen} = |\mathrm{Index}(V_{\rm BHOP})|/|\Gamma| = 27/9 = 3$. The full heterotic Bianchi anomaly cancellation $c_2(TX) - c_2(V_{\rm BHOP}) - c_2(V'_{\rm BHOP}) = 6\,\tau_1^2$ (BHOP Eq. 99) is satisfied by a positive-tension 5-brane wrapping τ_1^2, and Mumford–Takemoto polystability is established on the documented Kähler region $\{t_a \in [0.25, 8.0],\ t_a/t_b \in [0.25, 4.0]\}$. Both checks are programmatically verified upstream of this paper at commits `a3317377` (anomaly cancellation) and `a4d1bc72` (polystability). The matter content is delivered by $H^1(X/\Gamma, V_{\rm BHOP})$ decomposing into three copies of the $\mathbf{16}$ of $SO(10)$ under the $\mathbb{Z}_3 \times \mathbb{Z}_3$ Wilson-line characters, with right-handed neutrinos arising automatically as the singlet $\mathbf{1}$ in $\mathbf{16} = \mathbf{10} + \bar{\mathbf{5}} + \mathbf{1}$. This commitment supersedes the wave-28-era rank-3 $SU(3)$ "non-standard" bundle placeholder, which was retracted in the audit at `p_stage3_bundle_physics_audit.md` (commit `0cad4b0b`) for failing the full Bianchi identity, the $N_{\rm gen} = 3$ Atiyah–Singer count (the rank-3 placeholder did not reproduce the BHOP-2005 Eq. 88 index of -27), polystability wiring, and the 3:3:3 Wilson-character decomposition of H^1. Alternative chains (the Donagi–Ovrut $SU(5)$-structure-group route, or $E_8 \supset SO(16)$) admit consistent compactifications but are not the framework's commitment. The framework's mass formulas use E_8 Coxeter-spectrum data and Wilson-line-broken chain assignments that are largely independent of which standard chain is selected; the per-particle Coxeter-step assignments listed in App. A are framework structural commitments that the BHOP $SU(4)$ identification anchors to a specific heterotic bundle.

e. Substrate-physical reading of the rank-4 internal symmetry (I, suggestive; not a BHOP-2005 commitment). The rank-4 $SU(4)$ here is the *bundle structure group* of $V_{\rm BHOP}$ acting on holomorphic sections of V on the cover, *not* the gauged $SU(4)_C$ of Pati–Salam 1974 [15] acting on Standard-Model fermion 4-tuples. The framework offers a *suggestive* substrate-physical reading in the spirit of Pati–Salam $SU(4)_C$ colour-lepton unification — the four internal modes of the bundle treated as three colour modes plus a fourth "lepton-number" mode — because both groups are $SU(4)$ and both relate to lepton/quark unification. However, the identification of the bundle structure group with a 4D gauged $SU(4)_C$ is a non-trivial *additional* structural step that BHOP-2005 §6 does not assume: BHOP's $SU(4)_{\rm bundle}$ acts on holomorphic sections, not on Standard-Model fermion fields, and the GUT chain $E_8 \to SO(10) \times SU(4)_{\rm bundle} \to SU(5) \times U(1) \to G_{\rm SM} \times U(1)_X$ has no Pati–Salam stage in the gauge-symmetry-breaking ladder. The substrate-physical mechanism realising any bundle→gauge identification is currently an open structural step. The framework's empirical content is independent of the identification: chiral generations descend from the $\mathbb{Z}_3 \times \mathbb{Z}_3$ Wilson characters acting on $H^1(X/\Gamma, V_{\rm BHOP})$ regardless of the Pati–Salam reading, and right-handed neutrinos are automatic from the $\mathbf{16}$ of $SO(10)$ (the singlet $\mathbf{1}$ in $\mathbf{16} = \mathbf{10} + \bar{\mathbf{5}} + \mathbf{1}$). The rank-4 commitment is preferred over the wave-28 rank-3 $SU(3)$ placeholder because the rank-4 extension bundle satisfies the BHOP-2005 Eq. 88 index of -27, full Bianchi anomaly closure, polystability, and the $\mathbf{16}$-decomposition required for right-handed neutrinos — not because the Pati–Salam reading is structurally forced. The residual $U(1)_X$ surviving Wilson-line breaking is anomalous and is broken by the standard heterotic Green–Schwarz mechanism (anomalous-$U(1)$ Stueckelberg coupling to the dilaton/ B-field universal axion); the substrate-physical mechanism for this GS cancellation is an open step.

B. Empirical CY3 substrate identification

The framework's substrate-physical forcing chain selects the Hodge $(3,3)$, $\pi_1 = \mathbb{Z}_3 \times \mathbb{Z}_3$ branch on which Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$ sits. The structurally-closest distinct

three-generation Calabi–Yau threefold satisfying $c_1 = 0$ and admitting a free finite-group action with Standard-Model-compatible Wilson breaking is the Tian–Yau quotient $\mathrm{TY}/\mathbb{Z}_3$ (Tian–Yau 1987 [16], Hodge $(h^{1,1}, h^{2,1}) = (6, 9)$, $\pi_1 = \mathbb{Z}_3$, $\chi = -6$). Tab. II records the structural distinction.

To convert this from a literature-driven choice into an empirical identification, we ran two independent physics-channel discriminators on the Schoen and Tian–Yau Donaldson-balanced metrics: a quark-chain channel and a lepton-chain channel. The pipeline as currently implemented computes, for each candidate, a Galerkin-residual sweep at successive test-degrees $k_{\mathrm{test}} \in \{2, 3, 4, 5\}$ of the discrepancy between the metric-Laplacian eigenvalues (projected onto the framework's $\sqrt{2}$-chain integer/half-integer positions for the quark channel and the φ-chain integer-exponent positions for the lepton channel) and the structural sub-Coxeter chain expected from the framework. The signed channel residual difference $\delta = \rho_{\mathrm{Schoen}} - \rho_{\mathrm{TY}}$ is interpreted as a per-channel $\ln \mathrm{BF}_{\mathrm{TY/Schoen}} = \delta/2$ under a Laplace/BIC heuristic. The combined channel report is $\ln \mathrm{BF}_{\mathrm{TY/Schoen}} = -14.76$ (Tab. III), corresponding to the highest tier of the Kass–Raftery scale [24] ("very strong" evidence at $|\ln \mathrm{BF}| > 5$; equivalently $\log_{10} \mathrm{BF} > 6$, well beyond Jeffreys's "decisive" threshold of $\log_{10} \mathrm{BF} > 2$ [26]), with posterior odds Schoen:TY $= 2.6 \times 10^6$. The Wilks/Cowan–Cranmer–Gross–Vitells χ^2_1 equivalence $n_\sigma = \sqrt{2|\ln \mathrm{BF}|}$ yields 5.43σ-equivalent discrimination as an order-of-magnitude non-nested heuristic.

a. Convergence status of the chain-channel sweep. The chain-channel discriminators reported here are *indicative, not converged* at the strict 10% basis-size convergence criterion. The current production sweep at $n_{\mathrm{pts}} = 25{,}000$, $k = 3$, $k_{\mathrm{test}} \in \{2, 3, 4, 5\}$ records the channel JSONs at `p7_11_quark_chain.json` and `p7_11_lepton_chain.json`, both of which carry `converged: false` flags with consecutive-degree relative deltas of order 11–35% on the Schoen side and the signed-delta δ residual. The $\ln \mathrm{BF} = -14.76$ figure is therefore a preliminary discriminator, not a converged Bayes factor in the strict statistical sense; we report it as supporting evidence alongside the catalogue uniqueness sweep (which *is* closed modulo the gCICY non-manifest residue) and the Tian–Yau Yukawa structural exclusion (which *is* closed at the searched bidegree range). The full replacement of the metric-Laplacian chain-residual likelihood by a converged PDG-fermion-mass Yukawa-cohomology pipeline (Anderson–Karigiannis–Larfors–Lukas–Schneider style: HYM Hermitian metric on V, harmonic representatives of $H^1(X/\Gamma, V)$, Yukawa cohomology pairing, Kähler normalisation, two-loop SM RG running) is the in-flight remediation specified in the project's `REMEDIATION_PLAN_OPTION_B.md`, Phase 2; we expect to retract or upgrade the present chain-channel BF figure on its completion. The catalogue uniqueness and TY-exclusion arguments do not depend on the chain-channel convergence and stand on their own.

b. σ-channel as discriminability, not BF input. The L^2 Donaldson §3.4 weighted-variance Monge–Ampère residual σ on the Donaldson-balanced metric (Donaldson 2009 [25], PAMQ **5**, p. 587, eq. 3.4: $\sigma^2 = \langle(\eta/\kappa - 1)^2\rangle_w$ with $\kappa = \langle\eta\rangle_w$, matching the production implementation in `schoen_metric.rs::compute_sigma` and `ty_metric.rs::compute_sigma`) is a separately-computed statistical discriminator that distinguishes the two candidates' metrics with $|t| = 6.92$ at $k = 3$, BCa $[5.30, 9.04]$ (Tab. IV). The k-scan across $k = 3, 4, 5$ reveals the empirical signature of basis-size dominance: as Schoen's basis dimension grows monotonically ($27 \to 48 \to 75$), the strict-converged Schoen subset shrinks ($16/20 \to 7/20 \to 6/20$) and the heavy-tailed σ_{Schoen} distribution increasingly drives the $|t|$-statistic. To test whether this is a real geometric signal or a basis-size artifact, we performed a paired-bootstrap test at matched basis size $n_b = n_b^{\mathrm{Schoen}} = 27$ between the two candidates' metrics, 20 seeds, $B = 10^4$ resamples (Tab. V). The signed gap *inverts* when bases are matched, with disjoint BCa CIs and bootstrap probability of opposite sign equal to unity, and is the empirical basis for excluding σ from the multi-channel Bayes factor (`for_combination = false` flag on the `ChannelEvidence` record). The matched-basis comparison is currently implemented as a lex-sorted truncation of the larger TY basis to Schoen's 27-monomial dimension; a geometric basis projection (eigenvalue-ranked principal-components of the Laplacian, or harmonic-form ranking) is the in-flight upgrade to the matched-basis defense (`REMEDIATION_PLAN_OPTION_B.md` Phase 3.3). The σ-discriminability is reported *only* as a separate $|t|$-statistic, not as a likelihood contribution; the Bayes-factor exclusion of σ is conditional on the matched-basis defense surviving the geometric-projection upgrade.

c. Tian–Yau Yukawa-channel structural exclusion. Beyond the chain-channel Bayesian preference, we attempted to construct a working three-generation Standard-Model bundle on $\mathrm{TY}/\mathbb{Z}_3$ in the bicubic CICY ambient $\mathbb{P}^3 \times \mathbb{P}^3$ at $h^{1,1} = 2$, mirroring the working three-factor monad construction we use on Schoen (Tab. VII). A nine-cycle hypothesis-test enumeration over rank-3 SU(3) line-bundle-sum and monad bundles, summarised in Tab. VI, found no candidate that simultaneously satisfies $c_1(V) = 0$, $\Sigma\, c_3(V) = \pm 18$, $\mathrm{ch}_2(V) \leq \mathrm{ch}_2(TX)$, slope-stability via line-destabiliser test, balanced 1:1:1 Wilson $\mathbb{Z}_3$ phase partition, basis-rank ≥ 6 for three-generation Yukawa population, and Mumford–Takemoto SU(3) stability [23]. The line-bundle-sum result reproduces and extends Anderson–Gray–Lukas–Palti [17] §5.3's catalogue exclusion of $h^{1,1} \in \{2, 3\}$ to broader bundle constructions and to the explicit Mumford–Takemoto stability check. The leading rank-7 monad survivor of the topological + summand-stability constraints, $V_{\min 2}$, is mathematically falsified by $h^0(V_{\min 2}) = \mathbb{C}^2 \neq 0$ in the long exact sequence of $V = \ker(B \to C)$, violating the $h^0(V) = 0$ requirement for stable rank-3 SU(3) bundles on a Calabi–Yau threefold (cf. [23] §1.2). This structural

TABLE II. Structural comparison of the substrate's predicted Calabi–Yau threefold (Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$, Donagi–Ovrut–Pantev–Reinbacher [12]) against the closest distinct three-generation competitor (Tian–Yau $\mathbb{Z}_3$). The framework's substrate-physical forcing chain (self-conjugacy on the internal Hodge decomposition, two-stage Wilson breaking $E_8 \to G_{\mathrm{SM}}$, Atiyah–Singer three-generation index) selects the Hodge $(3,3)$ / $\pi_1 = \mathbb{Z}_3 \times \mathbb{Z}_3$ branch on which Schoen sits and Tian–Yau does not.

Quantity	Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$	Tian–Yau $\mathbb{Z}_3$	Selecting hypothesis
$(h^{1,1}, h^{2,1})$	$(3,3)$	$(6,9)$	self-conjugacy + 3 generations
π_1	$\mathbb{Z}_3 \times \mathbb{Z}_3$	$\mathbb{Z}_3$	two-stage Wilson breaking
$\chi(X)$	0	-6	deformation-class distinction
$2\,\omega_{\mathrm{fix}}$	246/248	245/248	gateway-formula ratio

TABLE III. Multi-channel Bayes factor for the substrate-CY3 hypothesis test, $k = 3$, $n_{\mathrm{pts}} = 25{,}000$ chain / 40,000 σ, 20 seeds. Sign convention: $\ln \mathrm{BF} > 0 \Leftrightarrow$ Tian–Yau favoured, $< 0 \Leftrightarrow$ Schoen favoured. The combined chain-channel result clears decisive Bayesian strength on the Kass–Raftery scale ($|\ln \mathrm{BF}| > 5$). The σ-channel is reported separately as discriminability and is excluded from the model-comparison Bayes factor (rationale and empirical defence: Tab. V).

Channel	Likelihood	$\ln \mathrm{BF}_{\mathrm{TY/Schoen}}$	n_σ-equivalent
$\mathrm{chain}_{\mathrm{quark}}$ ($\{m_b, m_s, m_d\}$)	log-χ^2 to PDG	-5.60	—
$\mathrm{chain}_{\mathrm{lepton}}$ ($\{m_\tau, m_\mu, m_e\}$)	log-χ^2 to PDG	-9.16	—
Combined (physics preference)	—	$\mathbf{-14.76}$	$\mathbf{5.43\sigma}$
Strength (Kass–Raftery)	—	**Strong (Schoen-favoured)**	
Posterior odds Schoen:TY	—	$2.6 \times 10^6 : 1$	
σ-channel (discriminability only; *excluded from BF*)	Welch t on σ-distribution	—	6.92σ

exclusion is independent of the Bayesian-discrimination evidence and converges on the same conclusion via different mathematical machinery.

d. Catalogue uniqueness at $h^{1,1} = h^{2,1} = 3$. The chain-channel Bayes factor and the Tian–Yau Yukawa exclusion identify Schoen as the favoured candidate within the framework's pre-narrowed two-element shortlist. To upgrade this from a pairwise discrimination to a catalogue-level uniqueness statement we performed a four-cycle sweep of the published smooth-CY3 universe at $(h^{1,1}, h^{2,1}) = (3,3)$ for non-Schoen entries admitting a free $\mathbb{Z}_3 \times \mathbb{Z}_3$ action with Standard-Model-compatible Wilson-line breaking. Cycle 1 (Hodge filter) returned zero hard-match competitors across CICY (Anderson–He–Lukas [18]), Schoen-class fiber products, Constantin–Lukas free-quotient catalogue, Borisov–Caldararu Pfaffian threefolds, and gCICY (Anderson–Apruzzi–Gao–Gray–Lee [20]). Cycle 2 (free-action structural filter) returned zero non-Schoen survivors. Cycle 2.5 (computer-algebra sweep) verified by direct query against the Kreuzer–Skarke 473,800,776-polytope catalogue [19] that the $(3,3)$ Hodge bin is *literally empty* (response `#NF: 0` at output limits $L \in \{100, 1000, 10000\}$, cross-verified by sweeping the entire $\chi = 0$ self-mirror diagonal $h = 1, \ldots, 14$ and confirming the bin is empty for $h \leq 13$ and populated from $h = 14$). Cycle 2.6 (gCICY non-manifest Aut residue) reduced the last open cell by programmatic literature search of arXiv:1507.03235 for the literal $(3,3)$ Hodge pair (zero hits), cross-checked against Larfors–Schneider [21] and Constantin–Lukas–Mishra [22]; the strict-CAS portion of cycle 2.6 (computing the full ambient automorphism group via Sage / Macaulay2 / polymake and checking which $\mathbb{Z}_3 \times \mathbb{Z}_3$ subgroups act freely on a generic gCICY $(3,3)$) remains **DEFERRED** pending a CAS-equipped re-run; see `p_schoen_uniqueness_cycle2_5.md` §3.4 and the in-flight Phase 6.2 of the remediation plan. The combined catalogue scan, modulo the gCICY non-manifest CAS residue, supports the statement: *no published smooth Calabi–Yau threefold in the catalogued universe is established to satisfy the substrate-physical four-principle constraint at $(h^{1,1}, h^{2,1}) = (3,3)$ except Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$.* The remaining residual is a future gCICY codim $\geq (3,1)$ extension producing a $(3,3)$ entry, plus the deferred CAS check on existing gCICY $(3,3)$ classifications; both would require a re-test but do not affect the present empirical identification.

e. Net empirical statement. Three convergent lines of evidence support the empirical identification of the substrate's internal Calabi–Yau threefold as Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$ over the closest structural competitor Tian–Yau $\mathbb{Z}_3$. **Closed:** (a) the four-cycle catalogue uniqueness sweep finds no non-Schoen $(h^{1,1}, h^{2,1}) = (3,3)$ smooth CY3 with a free $\mathbb{Z}_3 \times \mathbb{Z}_3$ action across the published universe (CICY, Kreuzer–Skarke toric, Schoen-class, Constantin–Lukas, Borisov–Caldararu Pfaffian, gCICY); the Kreuzer–Skarke direct query records `#NF: 0` at output limits up to $L = 10{,}000$, cross-verified by sweeping the entire $\chi = 0$ self-mirror diagonal $h \in \{1, \ldots, 14\}$. (b) the Tian–Yau Yukawa channel is structurally excluded by a nine-cycle hypothesis-test enumeration showing that no rank-3 SU(3) stable polystable monad bundle on $\mathrm{TY}/\mathbb{Z}_3$ in the bicubic ambient at $h^{1,1} = 2$ in the searched bidegree range simultaneously satisfies all required physical constraints. **Indicative, not converged:** (c) the chain-channel Bayes factor at $\ln \mathrm{BF} = -14.76$ (posterior odds

TABLE IV. σ-channel discriminability k-scan on TY-vs-Schoen, Donaldson balanced metric (no Adam refinement), $n_{\text{pts}} =$ 40,000, 20 seeds, BCa-95% CI from $B = 10^4$ bootstrap resamples (DiCiccio–Efron 1996). Strict-converged tier: residual $< 10^{-6}$ AND iters $<$ cap, no Tukey trim, no guard-snapshot inflation. The $k = 3$ row is the publication discriminability headline; $k = 4, 5$ are reported as *supporting tiers at higher basis order*, not as parallel publication results. The monotone shrinkage of the strict-converged Schoen subset ($16/20 \to 7/20 \to 6/20$) as Schoen's basis dimension grows ($27 \to 48 \to 75$) is the empirical signature of σ as a basis-size-saturated indicator at finite k, motivating exclusion from the model-comparison Bayes factor (Tab. V).

k	$\langle\sigma_{\text{TY}}\rangle$	$\langle\sigma_{\text{Schoen}}\rangle$	$\lvert t\rvert$-stat	Tier-0 BCa 95% CI	$n^{\text{strict}}_{\text{Schoen}}/n^{\text{Schoen}}_{\text{basis}}$
3 (publication)	0.272	5.59	$\mathbf{6.92}\sigma$	$[\mathbf{5.30},\ \mathbf{9.04}]$	16/20 / 27
4 (damped, supporting)	0.278	13.21	3.82σ	[2.48, 6.37]	7/20 / 48
5 (supporting)	0.295	13.05	10.87σ	[8.79, 15.01]	6/20 / 75

TABLE V. Empirical defence of the σ-channel exclusion from the multi-channel Bayes factor: matched-basis-size paired bootstrap test on TY-vs-Schoen, $k = 3$, $n_{\text{pts}} =$ 40,000, donaldson_iters= 100, donaldson_tol= 10^{-6}, 20 seeds, $B = 10^4$ resamples, boot_seed $\in \{12345, 999, 31415\}$ for jitter check. At native basis sizes $\sigma_{\text{TY}} < \sigma_{\text{Schoen}}$ as expected. **When the two metrics' bases are truncated to a common dimension $n_b = 27$ (Schoen's native value), the sign of the σ-gap inverts**, with disjoint BCa CIs and paired-bootstrap probability of opposite sign equal to unity over 10^4 resamples. The artifact magnitude is $\sim 3\times$ the native gap, in the opposite direction. This is the textbook signature of a basis-size-induced artifact, not a likelihood contribution, and is the empirical basis for excluding σ from the model-comparison Bayes factor of Tab. III.

Quantity	Mean	BCa 95% CI
$\Delta\sigma_{\text{native}}$ ($n_{\text{TY}} = 87,\ n_{\text{Schoen}} = 27$)	-7.89	$[-15.25,\ -5.52]$
$\Delta\sigma_{\text{matched}}$ ($n_{\text{TY}} = n_{\text{Schoen}} = 27$)	$\mathbf{+23.59}$	$[\mathbf{+11.33},\ \mathbf{+42.66}]$
$\lvert\Delta_{\text{matched}}\rvert/\lvert\Delta_{\text{native}}\rvert$	2.99	[1.54, 6.55]
$\mathbb{P}$(opposite sign | boot)	**1.0000** (10^4 resamples)	
Bootstrap-seed jitter on BCa-low	0.52 (range over $\{12345, 999, 31415\}$)	

2.6×10^6:1 Schoen:TY; nested-LRT σ-equivalent 5.43σ) is computed from a metric-Laplacian Galerkin-residual sweep that has not yet met the strict 10% basis-size convergence criterion at $k_{\text{test}} \in \{2, 3, 4, 5\}$; we report it as supporting evidence and expect to retract or upgrade the figure on completion of the full PDG-fermion-mass Yukawa-cohomology pipeline (Phase 2 of the remediation plan). The Donaldson σ-channel is excluded from the model-comparison Bayes factor by a paired-bootstrap matched-basis sign-reversal test (currently using lex truncation; geometric-projection upgrade is in flight at Phase 3). *Conditional on the open methodological items below and the in-flight pipeline upgrades enumerated in Sec. XV, the substrate-physical four-principle constraint narrows the candidate manifold to a single element of the catalogued universe at the searched Hodge bin.*

f. What this section does not *claim.* We are explicit about the residual scope. (i) We do *not* claim Schoen-uniqueness in full generality across the Calabi–Yau threefold landscape; the catalogue uniqueness statement is at the single Hodge bin $(h^{1,1}, h^{2,1}) = (3, 3)$ and is conditional on the gCICY non-manifest CAS residue (item iii below). (ii) The chain-channel Bayes factor $\ln \text{BF} = -14.76$ is *not* a converged result at the strict 10% basis-size convergence criterion: both the quark-chain and lepton-chain JSON outputs (`p7_11_quark_chain.json`, `p7_11_lepton_chain.json`) carry `converged: false`. The figure is reported as indicative; the full PDG-fermion-mass Yukawa-cohomology pipeline replacement is in-flight Phase 2 of the remediation plan. (iii) The gCICY $(3, 3)$ uniqueness check has **deferred** the strict computer-algebra-system portion of cycle 2.6 (Sage / Macaulay2 / polymake-based ambient automorphism enumeration) pending CAS environment availability; the literature-cross-check portion is closed. A future gCICY codim $\geq (3, 1)$ extension producing a $(3, 3)$ entry would also require a re-test. (iv) We do *not* claim that the Bayesian discrimination is independent of the Donaldson convention adopted for the Schoen and Tian–Yau metrics; the σ-channel result is sensitive to h-inversion handling and Kähler-form projection, and we report only the conventions used here. The σ-channel matched-basis defense currently uses lex truncation rather than a geometric basis projection; the geometric-projection upgrade and the full Donaldson convention robustness sweep are in-flight Phase 3 of the remediation plan. (v) The TY Yukawa exclusion is at the searched bidegree range in the bicubic ambient and is consistent with, but does not rigorously imply, the absence of any TY-supported three-generation bundle outside that range. (vi) The bundle data we use on Schoen, and the line-bundle enumerations cited from the literature, are subject to the bundle-provenance audit summarised in Sec. XV. None of items (i)–(vi) affects the numerical computations of Secs. III–XI, which proceed from the Schoen identification taken as a starting commitment.

TABLE VI. Structural exclusion of a working three-generation Standard-Model bundle on the closest competitor manifold Tian–Yau $\mathbb{Z}_3$ (bicubic CICY in $\mathbb{P}^3 \times \mathbb{P}^3$, $h^{1,1} = 2$): nine-cycle empirical enumeration. Constraints applied per cycle, in chain: $c_1(V) = 0$ AND $\Sigma\, c_3(V) = \pm 18$ for three generations downstairs after the $\mathbb{Z}_3$ quotient AND $\mathrm{ch}_2(V) \leq \mathrm{ch}_2(TX)$ for anomaly cancellation AND slope-stability via line-destabiliser test on the Kähler cone AND balanced 1:1:1 Wilson $\mathbb{Z}_3$ phase partition on V AND basis-rank ≥ 6 for three-generation Yukawa population AND Mumford–Takemoto SU(3) stability (Huybrechts–Lehn 2010 §1.2: $h^0(V) = 0$). Cycles run on monad bundles $V = \ker(B \to C)$ with B, C line-bundle sums, bidegrees in $[-2, 2]^2$. The cycle 1 line-bundle-sum exclusion empirically reproduces and extends Anderson–Gray–Lukas–Palti (arXiv:1202.1757) §5.3's published result that no phenomenologically viable model exists at $h^{1,1} \in \{2, 3\}$ in their scan. No candidate survives all nine constraints simultaneously.

Cycle	Hypothesis (rank-3 SU(3) bundle on TY/$\mathbb{Z}_3$ bicubic)	Failure mode	Survivors
H1	line-bundle sums, Wilson 1:1:1, $c_3 = \pm 18$	polystability fails universally	$0/{\sim}5.97 \times 10^6$
H2 v1/v2	monad $\ker(B \to C)$, weak polystability filter	polystability under correct slope	$0/-$
H3	SU(4)/SO(10) rank-4 alternative embedding	same polystability obstruction	$0/{\sim}9 \times 10^4$
H2 v3	monad, closed-negative-cone polystability	(passes topological + summand-stable)	$10/{\sim}3.78 \times 10^6$
C4	$V_{\min}$ wired ($\mathrm{rk}\, B = 4$)	basis-rank deficit; 3 modes $\neq 9$	0 (operationally)
C5	rank-$B \geq 6$ rescan	(10 v3 candidates re-confirmed)	10
C6	$V_{\min 2}$ wired ($\mathrm{rk}\, B = 7$)	H^0-seed empty on neg. summands	0 (operationally)
C7	H^0/H^1 cohomology audit	(LES-collapse argument validates AKLP/Schoen)	—
C8	$V_{\min 2}$ mathematical falsification	$h^0(V) = \mathbb{C}^2 \neq 0$: violates Mumford–Takemoto	**0**
C9	$H^0(V) = 0$ filter on cycle-5 v3 set	operationally infertile: $\Sigma h^0(B) = \Sigma h^1(B) = 0$	**0/10**
Net empirical result: no rank-3 SU(3) Yukawa-active SM monad bundle in scanned bidegree range.			

TABLE VII. Working three-generation Standard-Model bundle on Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$ used by the chain channels of Tab. III. Three-factor monad construction $V = \ker(B \to C)$ on the Schoen ambient $\mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^1$. The Wilson $\mathbb{Z}_3 \times \mathbb{Z}_3$ phase partition splits the V-summands evenly into three classes (one per generation) under the projection $(a - b) \bmod 3$ on the bidegrees, breaking the previously degenerate sector assignment that affects Tian–Yau under the same construction (Tab. VI). Anomaly cancellation $\mathrm{ch}_2(V) \leq \mathrm{ch}_2(TX)$ is satisfied with the indicated 5-brane class.

Object	Specification	Property
B	$\mathcal{O}(1,0,0)^2 \oplus \mathcal{O}(0,1,0)^2 \oplus \mathcal{O}(0,0,1)^2$	rank 6
C	$\mathcal{O}(1,1,0) \oplus \mathcal{O}(0,1,1) \oplus \mathcal{O}(1,0,1)$	rank 3
$V = \ker(B \to C)$	rank 3 SU(3)	$c_1(V) = 0$, $\Sigma\, c_3(V) = \pm 18$
Wilson Z_3 partition on V	$(1,1,1)$ across phase classes	balanced; canonical sector assignment
Anomaly cancellation	$\mathrm{ch}_2(V) \leq \mathrm{ch}_2(TX)$ with $W = (27, 27)$	satisfied
Yukawa coverage on V	$Y_u : 9/9$, $Y_d : 9/9$ on h_0 slice	operationally productive

C. The E_8 Coxeter spectrum

The E_8 Lie algebra has dimension $\dim(E_8) = 248$, rank 8, 240 roots, Coxeter number $h(E_8) = 30$, and Weyl group order $|W(E_8)| = 696{,}729{,}600$. The exponents of E_8 are the eight integers

$$\{m_1, \ldots, m_8\} = \{1, 7, 11, 13, 17, 19, 23, 29\}, \qquad (3)$$

palindromically symmetric around $h/2 = 15$ in pairs $(1, 29)$, $(7, 23)$, $(11, 19)$, $(13, 17)$, satisfying $m_j + m_{9-j} = h$. The longest Weyl element acts as $w_0 = -1$ on the Cartan subalgebra of E_8 (a property shared by $A_1, B_n, C_n, D_{2n}, E_7, E_8, F_4, G_2$). The harmonic threshold $h/2 = 15$ is the framework's chirality-flip locus for the Coxeter chain and is not itself an exponent.

a. Framework extension beyond the standard Coxeter spectrum. The mass-formula construction in this paper parameterizes *continuous* chain positions $k \in \mathbb{R}_{\geq 0}$, not just the eight integer exponents above. Half-integer positions ($k_d = 6.5$, $k_W = 34.5$, $k_t = 36.5$, etc.) and "cycle-2" positions ($k > h$) are framework-specific extensions that interpolate between integer exponents and re-use the periodicity of the Coxeter element across multiple cycles. These are structural assignments within the framework rather than standard Coxeter-theoretic invariants. The structural "boundary capacity" $h + m_{\max} = 59$ used in the dark-sector coupling family (Sec. V) is likewise a framework combination of E_8 invariants rather than an established Lie-algebra invariant in its own right.

D. The McKay correspondence

The McKay correspondence [5] establishes a bijection between finite subgroups of $SU(2)$ and the affine simply-laced ADE Dynkin diagrams: cyclic $\mathbb{Z}_n \leftrightarrow \widehat{A}_{n-1}$, binary dihedral $\leftrightarrow \widehat{D}_n$, binary tetrahedral $\leftrightarrow \widehat{E}_6$, binary octahedral $\leftrightarrow \widehat{E}_7$, and **binary icosahedral** $\leftrightarrow \widehat{E}_8$. The McKay correspondence itself identifies vertices of the affine $\widehat{E}_8$ Dynkin diagram with irreducible representations of the binary icosahedral group; it does *not*, on its own, single out a particular integer. Separately, the icosahedral

TABLE VIII. Methodological provenance and replicability conventions for the CY3-substrate-identification analysis (Tabs. II–VII). Run-time per sweep: ~ 2.5 h CPU on the basis-convergence sign-reversal test; ~ 3.5 h GPU on each k-scan production run; total $\sim$50 GPU-hours plus $\sim$30 CPU-hours across all reported tiers. The σ-discriminator throughout this paper is the weighted-L^2 Donaldson 2009 §3.4 functional, matching the production implementation; see row below.

Item	Setting / source
σ-functional definition	weighted L^2 Monge–Ampère residual: $\sigma^2 = \langle(\eta/\kappa - 1)^2\rangle_w$, $\kappa = \langle\eta\rangle_w$ (Donaldson 2009 §3.4 eq. 3.4)
σ-functional implementation	`schoen_metric.rs::compute_sigma`, `ty_metric.rs::compute_sigma` (CPU + GPU bit-exact mirror)
Bayes-factor sign convention	$\ln \mathrm{BF}_{12} > 0 \Leftrightarrow M_1$ favoured (Jeffreys/Kass–Raftery)
Strict-converged tier gate	residual $<$ tol AND iters $<$ cap, no Tukey trim
Bootstrap CI methodology	percentile and bias-corrected accelerated, $B = 10^4$ resamples
Bootstrap-seed jitter check	boot_seed $\in \{12345, 999, 31415\}$, target $< 0.05\sigma$ on BCa-low
Donaldson update (damped, $k \geq 4$)	$h \leftarrow \alpha\, T(G)^{-1} + (1-\alpha)\, h_{\mathrm{old}}$, $\alpha = 0.5$
GPU/CPU bit-exactness (h-pair)	mirrored 256-lane tree reduction, iter-0 $\Delta < 4.4 \times 10^{-16}$
Reproducibility manifest	SHA-256 chained event log per run
n_σ-equivalent from $\ln \mathrm{BF}$	$n_\sigma = \sqrt{2\lvert\ln \mathrm{BF}\rvert}$ (Wilks/Cowan–Cranmer–Gross–Vitells)
Source code	`book/scripts/cy3_substrate_discrimination/rust_solver/`
Output JSON catalogue	`output/p{5_10,7_11,8_1,_basis_convergence}_*.json`
Research log (9-cycle TY exclusion)	`references/p_ty_bundle_research_log.md`
Consolidated audit document	`references/cy3_publication_summary.md`

rotation group A_5 contains elements of order 5 in the $(2,3,5)$-triangle group, with $\varphi = 2\cos(\pi/5) = (1+\sqrt{5})/2$ as the icosahedral angular ratio. The framework adopts this fivefold rotation order, paired with the binary icosahedral McKay-dual of E_8, as the seed integer for the φ-chain. The use of the integer 5 is therefore a framework structural commitment combining the McKay-dual identification with the icosahedral rotation order, not a forced output of the McKay correspondence.

a. Two harmonic chains used by the framework. The mass-formula construction below distinguishes a φ-chain (with φ^k primary harmonics, motivated by 5-fold icosahedral recursion at the binary-icosahedral McKay-dual of E_8) from a $\sqrt{2}$-chain (with $(\sqrt{2})^k$ primary harmonics, motivated by 4-fold octahedral recursion at the binary-octahedral McKay-dual of E_7, where $\sqrt{2} = 2\cos(\pi/4)$). Charged leptons and neutrinos populate the φ-chain; quarks and electroweak bosons populate the $\sqrt{2}$-chain. This sector split is a *framework structural assignment* rather than a derivation from E_8 alone — a single E_8 algebra does not naturally decompose into icosahedral and octahedral sub-recursions, and the allocation of leptons to φ and quarks to $\sqrt{2}$ is adopted on phenomenological grounds (matching observed masses with small-integer Coxeter-step assignments). A first-principles derivation of the chain assignment from a specific Calabi–Yau compactification is left for future work.

b. Chain factors are not Coxeter eigenvalues. We emphasise that the harmonic factors φ^k and $(\sqrt{2})^k$ that appear throughout the mass formulas below are *exponential growth/recursion factors* on a discrete chain — successive applications of a fixed scalar step — and are *not* eigenvalues of the Coxeter element of E_8. The latter are the unit-modulus roots of unity $|e^{2\pi i e_j/h}| = 1$ for $e_j \in \{1, 7, 11, 13, 17, 19, 23, 29\}$, which describe rotational structure on the Coxeter plane and have nothing directly to do with mass-ratio recursion. The exponents e_j enter the mass formulas separately, as integer positions used to label palindromic pairings (e.g. $11 + 19 = h$); the multiplicative factors $\varphi^k, (\sqrt{2})^k$ encode the geometric step magnitude of the chain itself. These are two mathematically distinct uses of E_8 data, and the framework's construction is the joint use of both.

E. Substrate-tension interpretation

The framework adopts a substrate-tension interpretation of the Higgs mechanism, motivated by Anderson's superconducting precursor to gauge-boson mass generation [7]; the resulting picture is similar in spirit to Volovik's quantum-liquid / helium-droplet emergent-vacuum program [8, 9], though arrived at independently and from a different structural starting point. Within the substrate framework, the Higgs vacuum expectation value v is interpreted as the baseline relaxed tension of the $E_8 \times E_8$ substrate, and massive SM particles are localized deformations of this tension at structurally-determined Coxeter positions. Photons are massless because they correspond to solitonic excitations of the substrate that do not deform its baseline tension.

a. Foundations deferred to companion paper. The substrate-physical reading of c, $\hbar$, the elastic-medium identification $c = \sqrt{\Theta/\mathcal{I}}$, and the macroscopic emergence of general relativity, quantum mechanics, Newton's laws of motion, and Maxwell's equations from the same elastic substrate that hosts the $E_8 \times E_8$ Coxeter structure are not load-bearing for the numerical demonstration of this paper, and the present paper takes them as part of the starting commitment of Table I. Their derivation chains are developed in the companion paper [43] and the foundational monograph [42]. We do not repeat them here.

III. GATEWAY FORMULA

The remainder of this paper is the direct computation of SM observables from the starting assumptions of Table I. There is no per-particle scheme choice and no fitted continuous parameter: every numerical value reported below is the framework's structural prediction. PDG-matching residuals are predictions versus measurement, not fits versus measurement.

The electron sits at the gateway $k = 0$, defined as one chain step before the first E_8 exponent (at position 1). The chain depth from gateway to harmonic threshold is

$$M = h/2 - 1 = 14. \tag{4}$$

The framework's effective coupling scale is

$$h_{\text{eff}} \;=\; \frac{m_e}{2\omega_{\text{fix}}} \;=\; m_e \cdot \frac{\dim}{\dim - 2} \;=\; m_e \cdot \frac{124}{123}, \tag{5}$$

where $\omega_{\text{fix}} = 1/2 - 1/\dim = 123/248$ is the gateway normalization factor. Equivalently, with the Planck mass $m_{\text{Planck}} = \sqrt{\hbar c/G}$ taken from the structurally predicted G of Sec. XI, $h_{\text{eff}} = F_e \cdot m_{\text{Planck}} \cdot 124/123$, with F_e fixed entirely by Eq. (6) below; the two readings agree at the 17 ppm residual reported here, and the downstream mass formulas are operationally identical under either reading.

The dimensionless ratio of the electron mass to the Planck mass is predicted by the framework as

$$\boxed{F_e \;=\; \frac{m_e}{m_{\text{Planck}}} \;=\; \frac{\dim - 2}{\dim} \cdot \frac{5}{\dim \cdot h^M} \cdot \frac{1}{1 - \frac{1}{4(\dim - 1)}}.} \tag{6}$$

The right-hand side contains *no measured input*: every quantity is a structural integer of the starting assumptions (Table I). This is a zero-parameter prediction of the dimensionless electron-to-Planck mass ratio.

Each factor has a specific structural role in the framework:

- Gateway factor $(\dim - 2)/\dim = 246/248$: framework dual-sector count subtracting two distinguished endpoints (one in each E_8).
- McKay icosahedral 5: framework-selected icosahedral order motivated by the binary-icosahedral McKay dual of E_8.
- Coxeter chain $h^M = 30^{14} \approx 4.78 \times 10^{20}$: each step contributes a factor of $h = 30$.
- Sub-harmonic loss $988/987 = 1/(1 - 1/988)$: the geometric sum $\sum_{n\geq 0}(1/988)^n$ of a per-step leakage rate $1/(4(\dim - 1)) = 1/988$ to all orders. The factor of 4 in $4(\dim - 1)$ admits a dual-sector / two-orientation reading ("2×2": two E_8 factors, each carrying both an upward and a downward chain-step orientation; $\dim - 1$ removes the distinguished anchor mode in each sector); a substrate-physical derivation of the 4 from a heterotic-anomaly index or a $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient trace on the Schoen threefold is part of the open companion-paper program (Sec. XV).

Numerically, $F_e = 4.18547 \times 10^{-23}$. The measured value $m_e^{\text{PDG}}/m_{\text{Planck}}^{\text{CODATA}}$ agrees with this structural prediction at 17 ppm. Newton's constant is the zero-parameter consequence $G = F_e^2 \cdot \hbar c/m_e^2$ developed in Sec. XI.

IV. CHARGED-FERMION MASSES

A. Mass formula schema

Every charged-fermion mass in the framework takes the unified form

$$\boxed{m_X \;=\; h_{\text{eff}} \cdot R_X(k_X, \text{sector}, \{\text{corr.}\}),} \tag{7}$$

where $h_{\text{eff}} = m_e \cdot \dim/(\dim - 2) = 124 m_e/123$ is the gateway-normalized scale (Sec. III), k_X is the particle's Coxeter step assignment in the E_8 spectrum (cf. App. A), and R_X is a structural ratio built from E_8 invariants $\{\dim, h, M, \text{rank}, m_{\max}\}$, the gateway-mixing parameter $\alpha_\tau = 1/(\dim + h) = 1/278$, and sector-specific sub-harmonic corrections. Numerical values of R_X for every particle are tabulated in Sec. XIII (Table XIII); the closed-form expressions are given below sector by sector.

B. Lepton sector

The electron, muon, and tau occupy integer E_8 exponents on the φ-chain (icosahedral bulk recursion) at $k_e = 0$ (gateway), $k_\mu = 11$, and $k_\tau = 17$ respectively. Their explicit closed-form ratios are

$$R_e \;=\; \frac{\dim - 2}{\dim} \;=\; \frac{246}{248}, \tag{8}$$

$$R_\mu \;=\; \varphi^{11} \cdot \big(1 + \Delta_\varphi\big) \cdot \big(1 - \alpha_\mu\big), \tag{9}$$

$$R_\tau \;=\; \varphi^{17} \cdot \big(1 - \Delta_\varphi\big) \cdot \big(1 - \alpha_\tau^{\text{full}}\big), \tag{10}$$

where the icosahedral-bulk closure factor is

$$\Delta_\varphi \;=\; \frac{1}{h + \varphi^2}, \tag{11}$$

and the sub-harmonic corrections are

$$\alpha_\mu \;=\; \sum_{n=1}^{\infty} c_n^{(\mu)} \alpha_\tau^n \;=\; 2\alpha_\tau^2 + \mathcal{O}(\alpha_\tau^3), \tag{12}$$

$$\alpha_\tau^{\text{full}} \;=\; \frac{\alpha_\tau}{1 + \varphi^2 \alpha_\tau} \;=\; \frac{1}{\dim + h + \varphi^2}. \tag{13}$$

Each lepton ratio has three structural pieces: a φ-chain primary harmonic φ^{k_ℓ} (icosahedral bulk recursion at the lepton's Coxeter exponent), an icosahedral-bulk closure

$(1 \pm \Delta_\varphi)$ with sign opposite for muon and tau (reflecting their opposite displacement from the harmonic threshold $h/2 = 15$), and a sub-harmonic series correction. The muon coefficient $c_2^{(\mu)} = 2$ is the framework's minimal assignment to the palindromic pair $(11, 19)$ centered on $h/2$. The tau correction $\alpha_\tau^{\text{full}}$ is the closed-form sum of the geometric series $\sum_n (-\varphi^2 \alpha_\tau)^n \alpha_\tau$ in the icosahedral self-coupling. Predicted (electron-anchor): $m_\mu = 105.6583756$ MeV (PDG match: +1.1 ppb), $m_\tau = 1776.8597$ MeV (-0.15 ppm).

C. Quark sector

Quarks occupy positions on the $\sqrt{2}$-chain (octahedral bulk recursion) at integer and half-integer Coxeter steps. Their explicit closed-form ratios are

$$R_q \;=\; (\sqrt{2})^{k_q} \cdot \big(1 + \chi_q \Delta_2\big) \cdot \big(1 + \lambda_q \tfrac{|k_q - h/2|}{2\,\dim}\big) \cdot \big(1 - \alpha_q\big), \quad (14)$$

where the octahedral-bulk closure factor is

$$\Delta_2 \;=\; \frac{1}{32}, \quad (15)$$

and $\chi_q, \lambda_q \in \{-1, 0, +1\}$ are per-quark chirality and per-step-loss signs respectively. The per-quark gateway-mixing sub-harmonic series α_q is non-zero only for the top quark, the only cycle-2 quark in the framework. For all cycle-1 quarks $\alpha_q = 0$; for the top quark,

$$\alpha_t \;=\; \frac{c_2^{(t)} \alpha_\tau^2}{1 + \alpha_\tau / h_{SU(3)}}, \quad c_2^{(t)} = \left(\frac{h}{h_{SU(3)}}\right)^2 = 100, \quad (16)$$

with $h_{SU(3)} = 3$ the colour Coxeter number. The leading coefficient $c_2^{(t)} = (h_{E_8}/h_{SU(3)})^2$ is the topological-subgroup ratio between the substrate's full Coxeter period and the colour Coxeter period, squared: ten colour cycles fit inside one full substrate cycle, and the top – the only cycle-2 colour-charged fermion – projects through this ratio twice (once for its colour-charge winding, once for its cycle-2 chain-position projection). The closure denominator $h_{SU(3)} = 3$ is the same colour-projected closure that appears in α_μ and α_H throughout the framework. For the strange quark sitting exactly at the harmonic threshold $k_s = h/2 = 15$, the per-step-loss factor $|k_s - h/2|/(2\,\dim) = 0$ regardless of the sign of λ_s, so λ_s is indeterminate (we record it as $\lambda_s = 0$ purely as bookkeeping notation). The chirality factor $\chi_s = 0$, in contrast, is a *distinct structural assignment*: it forces the closure factor $(1 + \chi_s \Delta_2) = 1$ at the chirality-flip locus and is the only non-trivial sign choice the strange quark makes:

The denominator $2\,\dim = 496$ in the per-step-loss term is the framework's chosen normalization by the full $E_8 \times E_8$ algebra dimension (visible plus dark sectors). All six quark masses lie within current PDG 2024 uncertainty bands; the worst central-value residual is +5618 ppm on the up quark, well within the ∼25% experimental uncertainty on the $\overline{\text{MS}}$ light-quark mass at $\mu = 2$ GeV.

TABLE IX. Per-quark Coxeter step k_q, chirality sign χ_q, and per-step-loss sign λ_q in the quark mass formula Eq. (14).

Quark	k_q	χ_q	λ_q	Cycle
up	4	+1	+1	1
down	6.5	−1	−1	1
strange	15	0	0	1 (threshold)
charm	22.5	+1	−1	1
bottom	26	−1	+1	1
top	36.5	+1	+1	2 (cycle 2)

D. Term-by-term decomposition: charged fermions

Each charged-fermion mass function decomposes into four contributions: the *primary harmonic* (Coxeter step magnitude), the *sub-harmonic correction* (recursive coupling at order α_τ^k), the *sector modifier* (chain-type factor: φ-chain or $\sqrt{2}$-chain), and the *dark-sector leakage* (boundary correction for cycle-2 modes only). For cycle-1 fermions the dark-sector leakage is absent. The decomposition for each particle:

a. Electron ($k_e = 0$).

$$m_e \;=\; h_{\text{eff}} \cdot \underbrace{\frac{\dim - 2}{\dim}}_{\text{gateway factor}}$$

- *Primary harmonic:* the gateway-factor ratio $(\dim - 2)/\dim = 246/248$ is the leading contribution. The numerator $\dim - 2 = 246$ subtracts the two dual-sector endpoints (one in each E_8); the denominator normalizes to the full algebra dimension.
- *Sub-harmonic correction:* none; the electron is the gateway reference and absorbs all sub-harmonic structure into the definition $h_{\text{eff}} = m_e \cdot \dim/(\dim - 2) = 124 m_e/123$.
- *Sector modifier:* gateway position (no chain-type distinction).
- *Dark-sector leakage:* not applicable (cycle-1 mode at $k = 0$).

b. Muon ($k_\mu = 11$), $R_\mu \approx 205.10$.

- *Primary harmonic:* φ^{11}, the icosahedral bulk-recursion contribution at the muon's φ-chain position $k_\mu = 11$ (the third E_8 exponent, on the Φ-below-threshold side of the palindromic spectrum).
- *Icosahedral-bulk closure:* $(1 + \Delta_\varphi)$ with $\Delta_\varphi = 1/(h + \varphi^2)$. The positive sign reflects the muon's position below the harmonic threshold $h/2 = 15$.
- *Sub-harmonic correction:* $(1 - \alpha_\mu)$ at order α_τ^2 with leading coefficient $c_2^{(\mu)} = 2$, assigned to the palindromic pair $(11, 19)$ centered on $h/2$.
- *Sector modifier:* φ-chain (icosahedral, lepton).
- *Dark-sector leakage:* not applicable (cycle-1 mode).

c. Tau ($k_\tau = 17$), $R_\tau \approx 3449.19$.

- *Primary harmonic:* φ^{17}, the icosahedral bulk-recursion contribution at the tau's φ-chain position $k_\tau = 17$ (the fourth E_8 exponent, the first above-the-threshold exponent).
- *Icosahedral-bulk closure:* $(1 - \Delta_\varphi)$ with $\Delta_\varphi = 1/(h + \varphi^2)$. The negative sign reflects the tau's position above the harmonic threshold $h/2 = 15$, opposite to the muon.
- *Sub-harmonic correction:* $(1 - \alpha_\tau^{\text{full}})$ where $\alpha_\tau^{\text{full}} = \alpha_\tau/(1+\varphi^2\alpha_\tau) = 1/(\dim + h + \varphi^2)$, the closed-form sum of the geometric series in the icosahedral self-coupling φ^2.
- *Sector modifier:* φ-chain (icosahedral, lepton).
- *Dark-sector leakage:* not applicable.

d. Quarks (six entries via Eq. (14)). The unified quark formula $R_q = (\sqrt{2})^{k_q} \cdot (1 + \chi_q\Delta_2) \cdot (1 + \lambda_q|k_q - h/2|/(2\dim))$ has three structural pieces common to all six quarks, with per-quark signs χ_q and λ_q from Table IX:

- *Primary harmonic:* $(\sqrt{2})^{k_q}$, the octahedral bulk-recursion contribution at the quark's Coxeter step.
- *Octahedral-bulk closure:* $(1 + \chi_q\Delta_2)$ with $\Delta_2 = 1/32$. The chirality sign χ_q alternates across the spectrum: $+1$ for "up-type" quarks (up, charm, top) and -1 for "down-type" quarks (down, bottom). The strange quark at the harmonic threshold has $\chi_s = 0$ (the closure factor collapses to 1 at the chirality-flip locus).
- *Per-step loss:* $(1 + \lambda_q|k_q - h/2|/(2\dim))$ with sign $\lambda_q \in \{-1, 0, +1\}$ per quark. The denominator $2\dim = 496$ is the framework's normalization by the full $E_8 \times E_8$ algebra dimension (visible plus dark sectors), reflecting the per-step leakage across both sectors as the chain advances away from the harmonic threshold $h/2$. The strange quark at $k_s = h/2$ has $|k_s - h/2| = 0$, so the loss factor is automatically unity for any value of λ_s; the entry $\lambda_s = 0$ in Table IX is a notational placeholder, not an independent structural assignment.
- *Sector modifier:* $\sqrt{2}$-chain at integer or half-integer position. The strange quark sits exactly at the harmonic threshold, the chirality-flip locus.
- *Dark-sector leakage:* the top quark is the only cycle-2 quark ($k_t = 36.5 > h = 30$); for the top, the $|k_t - h/2|$ distance is computed as written and the same formula Eq. (14) applies without an additional cycle-extension modifier.
- *Cycle-2 sub-harmonic correction (top quark only):* $\alpha_t = (h/h_{SU(3)})^2\alpha_\tau^2/(1 + \alpha_\tau/h_{SU(3)})$, with leading coefficient $(h/h_{SU(3)})^2 = 100$ – the topological subgroup ratio between the substrate's full Coxeter period $h_{E_8} = 30$ and the colour Coxeter period $h_{SU(3)} = 3$, squared (one factor for the top's colour-charge winding, one for its cycle-2 chain-position projection). Closure denominator $h_{SU(3)} = 3$. The other five quarks (cycle-1) have $\alpha_q = 0$ in Eq. (14); only the top, as the unique cycle-2 colour-charged matter mode, picks up the topological subgroup ratio in its sub-harmonic series.

V. HIGGS SECTOR AND DARK-SECTOR COUPLING FAMILY

A. Boson mass-formula schema and structural corrections

The electroweak boson masses and the Higgs vev occupy positions on the $\sqrt{2}$-chain (octahedral bulk recursion) in cycle 2 of the Coxeter spectrum. Their explicit closed-form ratios are

$$R_v = (\sqrt{2})^{k_v} \cdot \big(1 - \alpha_v\big), \tag{17}$$

$$R_H = \frac{(\sqrt{2})^{k_v}}{2} \cdot \big(1 + \delta_H\big) \cdot \big(1 - \alpha_H\big), \tag{18}$$

$$R_W = (\sqrt{2})^{34.5} \cdot \big(1 + \delta_W\big) \cdot \big(1 - \alpha_W\big), \tag{19}$$

$$m_Z = m_W/\cos\theta_W \quad \text{(SM tree-level)}, \tag{20}$$

where the Higgs-vev Coxeter position is

$$\begin{aligned} k_v &= h + \text{rank} - \text{rank}/h \\ &= 30 + 8 - 8/30 = \frac{1132}{30} \approx 37.733. \end{aligned} \tag{21}$$

The Higgs vev and Higgs boson share the same $\sqrt{2}$-chain primary-harmonic exponent k_v; the Higgs boson's amplitude is reduced by a factor of 2 (the breathing-mode amplitude relative to the substrate baseline tension), and acquires the $(1 + \delta_H)$ dark-sector leakage which the vev does not. The W boson sits at the half-integer cycle-2 position $k_W = 34.5$ on the $\sqrt{2}$-chain with leakage $(1+\delta_W)$ suppressed by an E_8 exponent factor of 19.

The boson-sector structural corrections are

$$\alpha_v = \frac{c_2^{(v)}\alpha_\tau^2}{1 + \alpha_\tau/(2\varphi)}, \quad c_2^{(v)} = \frac{\text{rank} \cdot h}{m_{\max}} = \frac{|\Phi_{E_8}|}{m_{\max}} = \frac{240}{29}, \tag{22}$$

$$\alpha_H = \frac{c_2^{(H)}\alpha_\tau^2}{1 + \alpha_\tau/h_{SU(3)}}, \quad c_2^{(H)} = \frac{(\text{rank} - 2) \cdot h}{m_{\max}} = \frac{180}{29}, \tag{23}$$

$$\alpha_W = \frac{c_2^{(W)}\alpha_\tau^2}{1 + \alpha_\tau/h_{SU(3)}}, \quad c_2^{(W)} = \frac{2}{e_2} = \frac{2}{7}, \tag{24}$$

$$\delta_H = \frac{1}{h + m_{\max}} = \frac{1}{59}, \tag{25}$$

$$\delta_W = \frac{\delta_H}{e_6} = \frac{1}{(h + m_{\max}) \cdot 19} = \frac{1}{1121}, \tag{26}$$

where $e_6 = 19$ (1-indexed in the ordered E_8 exponent list $\{e_1, \ldots, e_8\} = \{1, 7, 11, 13, 17, 19, 23, 29\}$) is the partner of the muon's E_8 exponent $e_3 = 11$ under the standard Coxeter duality $e_j + e_{\mathrm{rank}+1-j} = h$ [6]. For E_8, this duality groups the exponents into the four palindromic pairs $\{(1, 29), (7, 23), (11, 19), (13, 17)\}$, all summing to $h = 30$ and centered on $h/2 = 15$; the W-boson construction selects the pair $(e_3, e_6) = (11, 19)$ containing the muon's chain position.

The cycle-2 sub-harmonic series $\alpha_v, \alpha_H, \alpha_W$ (and α_t in the quark sector, Eq. (16)) share a common substrate-physical structure: each is a closed-form geometric-series sum of the form $c_2\alpha_\tau^2/(1 + \alpha_\tau/X)$, with mode-specific leading coefficient c_2 built from E_8 Coxeter slots (the eight Coxeter exponents) and the topological subgroup ratio $h/h_{SU(3)} = 10$, and closure denominator X from substrate- physical structural integers. The Higgs vev's closure denominator is 2φ (the icosahedral spin-doublet primary-harmonic step: McKay multiplicity 2 times the icosahedral recursion factor φ), reflecting the vev's role as the substrate baseline tension whose self-coupling is its own primary-harmonic step. The Higgs boson, W boson, and top quark closures are all $h_{SU(3)} = 3$ (the colour-projected closure), reflecting these modes' role as fluctuations on the colour-coupled cycle-2 vacuum. The leading coefficients are $c_2^{(v)} = \mathrm{rank}\cdot h/m_{\max}$ (full Cartan rank engages for the vev), $c_2^{(H)} = (\mathrm{rank}-2)\cdot h/m_{\max}$ (two Cartan directions lifted out for the breathing mode), $c_2^{(W)} = 2/e_2$ (vector multiplicity 2 over the substrate's second Coxeter slot), and $c_2^{(t)} = (h/h_{SU(3)})^2$ (topological subgroup ratio squared for the only cycle-2 colour-charged fermion). Each correction is built from framework-fixed E_8 invariants $\{\dim, h, \mathrm{rank}, m_{\max}, e_j\}$, the colour Coxeter number $h_{SU(3)} = 3$, and the gateway-mixing parameter α_τ. We note that the Higgs-vev coefficient $c_2^{(v)} = \mathrm{rank} \cdot h$ equals the order of the E_8 root system $|\Phi_{E_8}| = h \cdot \mathrm{rank} = 30 \cdot 8 = 240$, a standard Lie-algebraic identity for any irreducible finite Coxeter group [6]; the substitution $\mathrm{rank} \cdot h \to |\Phi_{E_8}|$ is the natural Lie-theoretic reading of α_v. The rank-and-rank-minus-two structures in α_v, α_H reflect a rank-locked-subspace projection shared with the dark-matter tower (Sec. X). *We use "rank-locked subspace" as framework-specific terminology*: it denotes the rank-rank = 8 invariant subspace that arises in the framework's bulk decomposition dim = dim −rank visible ⊕ rank-locked, distinct from any standard Cartan subalgebra construction. The phrase is defined operationally by its appearance in α_v, α_H, the dark-matter tower's $1 - \mathrm{rank}\,\alpha_\tau$ corrections, and certain quark sub-harmonics; a direct identification with a specific subspace of the Calabi–Yau cohomology is left for future work. The structural origin of δ_H and δ_W is motivated in the subsections below as a minimal closure rule compatible with the same geometric data.

The Higgs vev sits at cycle-2 chain position $k_v = 1132/30 \approx 37.733$, giving $v^{\mathrm{pred}} = 246.21965$ GeV (0.0001 ppm).

a. Closed-form structural identity for the vev. The chain-position prediction for the Higgs vev satisfies the substrate-physical identity at two levels of precision.

Leading identity (∼ 29 ppb residual):

$$\boxed{v \cdot (1 - \delta_W) \;=\; \dim(E_8) - 2 \;=\; 246 \;\; \text{(in GeV units)}} \tag{27}$$

to numerical residual ∼ 29 ppb (at the framework's gateway-mixing-tower depth-3, α_τ^3, scale). Equivalently,

$$\begin{aligned} v \;&=\; \frac{\dim(E_8) - 2}{1 - \delta_W} \;=\; \frac{246 \cdot 1121}{1120} \\ &=\; \frac{275766}{1120}\ \mathrm{GeV} \;\approx\; 246.21964\ \mathrm{GeV}, \end{aligned} \tag{28}$$

with $\delta_W = 1/(h + m_{\max})/e_6 = 1/1121$ the W boson's vector dark-sector leakage.

Extended identity (∼ 0.1 ppb residual, at floating-point precision):

$$v \cdot (1 - \delta_W) \;=\; (\dim(E_8) - 2) \cdot \left(1 + \frac{5}{8}\alpha_\tau^3\right), \tag{29}$$

with the additional gateway-mixing-tower depth-3 correction $\frac{5}{8}\alpha_\tau^3 \approx 29.1$ ppb. The leading factor 5/8 is the substrate-physical structural ratio

$$\begin{aligned} \frac{5}{8} \;&=\; \frac{\text{McKay icosahedral integer}}{\mathrm{rank}(E_8)} \\ &=\; \frac{\mathrm{rank}(E_8) - h_{SU(3)}}{\mathrm{rank}(E_8)} \;=\; \frac{8-3}{8}, \end{aligned} \tag{30}$$

"rank not consumed by colour, normalised by rank." The McKay icosahedral integer 5 that appears here is the same integer appearing in the gateway formula Eq. (6) as the icosahedral-seed factor.

Full structural ladder including dark-tower contributions. The substrate-physically complete form of the vev structural identity is a four-factor ladder:

$$v \;=\; (\dim -2) \cdot \frac{1}{1 - \delta_W} \cdot \left(1 + \frac{5}{8}\alpha_\tau^3\right) \cdot \big(1 + \zeta_{\mathrm{DM}}\big), \tag{31}$$

with the dark-matter tower's kinetic-mixing summed contribution

$$\zeta_{\mathrm{DM}} \;=\; \sum_{n=5}^{11} \epsilon_n \;=\; \sum_{n=5}^{11} \alpha_\tau^n \;=\; \alpha_\tau^5 \cdot \frac{1 - \alpha_\tau^7}{1 - \alpha_\tau} \;\approx\; 6.04\times10^{-13}. \tag{32}$$

Each factor in the ladder has clear substrate-physical content and operates at a distinct scale: $(\dim -2) \approx 246$ GeV is the gateway-saturated dimension count baseline; $1/(1 - \delta_W) \approx 1 + 893$ ppm is the macro-mode boundary coupling through the W-vector channel; $\frac{5}{8}\alpha_\tau^3 \approx 29$ ppb is the gateway-mixing-tower depth-3 correction; $\zeta_{\mathrm{DM}} \approx 0.6$ ppt is the dark-matter tower kinetic-mixing summed contribution. Each factor corresponds to one of the framework's three substrate-physical dark-sector coupling families: macro-mode boundary couplings (δ_W family), gateway-mixing-tower sub-harmonics (α_τ^n for $n \geq$

2), and dark-matter tower kinetic mixings ($\epsilon_n = \alpha_\tau^n$ for $n \geq 5$). The dark-tower factor $\zeta_{\rm DM}$ is structurally present and complete in the equation, but its magnitude is below PDG's precision floor for v (500 ppb) and below the framework's intrinsic comparison floor against PDG; including it does not improve the empirical match beyond the extended identity Eq. (29), but it represents the substrate-physically complete form. The same structural integer δ_W that governs the W boson's mass shift in Eq. (19) also governs the macro-mode elevation of the substrate's electroweak baseline: *one δ_W, two geometric contexts*, in the same pattern as the gateway formula's $(\dim -2)/\dim$ and the Weinberg angle's 2/9 each express the substrate gateway in different geometric slicings of the gauge phase-space.

Live anchor versus structural baseline. A direct empirical test against the high-precision electron and muon masses establishes that the substrate's mass spectrum is anchored to $v_{\rm observed}$, not to a counterfactual $v_{\rm natural} = \dim -2$. Anchoring the spectrum to $\dim -2$ predicts both leptons at -893 ppm against PDG (a $\sim 10^6\sigma$ deviation on the electron); anchoring to $v_{\rm observed}$ as in Eq. (28) predicts both within ~ 0.03 ppm. The substrate's chain-position couplings therefore reference the effective electroweak vacuum that has the dark-tower content already operating, not a counterfactual baseline that would exist without it. The structural identity Eq. (27) *expresses* the live anchor $v_{\rm observed}$ in closed form by factoring out the W-channel dark-tower elevation; it is not a separate, deeper anchor that the mass spectrum sees.

The Higgs vev is therefore the substrate's electroweak-baseline *macro-mode*: a cosmological-scale tension structure whose value in GeV units is fixed by the gateway-saturated dimension count of $E_8 \times E_8$ on Schoen $\mathbb{Z}_3 \times \mathbb{Z}_3$ combined with the W-channel dark-tower elevation that lifts it to the observed value. The framework's chain-position formula in Eq. (17) reproduces this identity through different but equivalent structural arithmetic, with numerical residual at the framework's typical sub-ppm precision floor.

B. Origin of the dark-sector coupling family

Particles whose Coxeter positions extend into cycle 2 of the E_8 spectrum (positions $k > h$) couple to the dark E_8 sector through the upper edge of the cycle-1 Coxeter spectrum. Substrate continuity across the dark/visible boundary forces a structural correction to their visible-sector mass formulas, equal to the inverse of the total Coxeter capacity available at the edge.

The structural "boundary capacity" of the E_8 Coxeter spectrum is the sum of the Coxeter number h and the largest exponent $m_{\rm max}$:

$$h + m_{\rm max} = 30 + 29 = 59. \tag{33}$$

This quantity uses two genuine E_8 invariants: h is the order of the Coxeter element (the period of the spectrum) and $m_{\rm max}$ is the highest non-trivial E_8 exponent (the upper exponent in the palindromic pair $(1, 29)$). Their sum is then a framework measure of the cycle-1 edge plus maximal-exponent displacement, not a standard Lie-algebra invariant in its own right.

C. The Higgs leak: structural closure rule for δ_H

The Higgs boson is the breathing-mode excitation of the substrate tension at $k_v - 2$ (one "inversion-step" below the vev). As a scalar mode-pattern, it couples isotropically to the boundary capacity at first order. A minimal scalar closure consistent with that boundary picture assigns the leading dark-sector correction

$$\boxed{\delta_H \;=\; \frac{1}{h + m_{\rm max}} \;=\; \frac{1}{59}} \tag{34}$$

($\approx 1.695\,\%$). The integer 1 in the numerator is the framework's minimal choice: a scalar mode has no internal index structure (no rank or exponent dependence) that would naturally supply an additional suppression factor. The factor $1/(h + m_{\rm max})$ then enters the Higgs-boson mass formula as the "boundary leakage" multiplicative correction at the breathing mode position $k_v - 2$, yielding $m_H^{\rm pred} = 125.19978$ GeV (1.75 ppm from PDG). This should be read as a structurally motivated closure rule awaiting derivation from the full compactification data, not as a completed CY3 cohomology computation.

D. The W leak: structural closure rule for δ_W

The W boson is a vector mode-pattern at cycle-2 Coxeter position $k_W = 34.5$, which lies on the $\sqrt{2}$ half-integer chain shared with the heavy-quark sector. As a vector mode, the W carries an internal index structure: its phase-alignment under $SU(2)$ projects onto a specific palindromic exponent pair on the φ-chain, namely the muon-paired pair $(11, 19)$ with $11 + 19 = h = 30$.

The corresponding vector closure rule suppresses the dark-sector boundary leakage for the W by the partner E_8 exponent across the palindromic pairing, $e_n = 19$ (the partner of 11, the muon's Coxeter position):

$$\boxed{\delta_W \;=\; \frac{\delta_H}{e_n} \;=\; \frac{1}{(h + m_{\rm max}) \cdot e_n} \;=\; \frac{1}{59 \cdot 19} \;=\; \frac{1}{1121}} \tag{35}$$

(≈ 892 ppm). Each factor in the denominator is fixed by E_8 structure once this closure pattern is adopted:

- $h = 30$ — Coxeter number (algebra-fixed).
- $m_{\rm max} = 29$ — largest E_8 exponent (algebra-fixed).
- $e_n = 19$ — partner exponent of the muon's position $k_\mu = 11$ under the palindromic pairing $m \leftrightarrow h - m$ centered on $h/2 = 15$ (algebra-fixed once the muon's Coxeter step is identified).

The ratio $\delta_W/\delta_H = 1/e_n = 1/19$ is therefore independently fixed by the E_8 exponent structure with no fitted continuous parameter; the same palindromic-pair structure that governs the muon-tau lepton mass relations governs the suppression of the W boson's dark-sector coupling relative to the Higgs's. Applying δ_W to the cycle-2 W position yields $m_W^{\rm pred} = 80.36950$ GeV (3.7 ppm from PDG). As with δ_H, this is best read as a framework closure rule sharpened by the E_8 exponent structure rather than as a fully first-principles compactification derivation.

E. Generalization to other cycle-2 modes

The same closure family extends to any cycle-2 Coxeter mode: a scalar in cycle 2 receives the bare δ_H correction; a vector or spinor in cycle 2 receives the suppressed correction $\delta_X = \delta_H/e_X$ where e_X is the E_8 exponent determined by the mode's internal index structure. The dark-sector coupling family is therefore graded by the E_8 exponent set $\{1, 7, 11, 13, 17, 19, 23, 29\}$, producing a discrete spectrum of structurally-fixed corrections rather than a continuous parameter. The Higgs self-coupling $\lambda_H = m_H^2/(2v^2)$ follows from the predicted m_H and v by the standard SM tree-level relation; its closed form and numerical value are presented in Sec. VI.

F. Term-by-term decomposition: electroweak bosons

The electroweak boson masses share the same four-component structure as the charged fermions (primary harmonic, sub-harmonic correction, sector modifier, dark-sector leakage), but all three massive bosons sit in cycle 2 of the Coxeter spectrum, so dark-sector leakage corrections are present for all of them.

a. Higgs vacuum expectation value ($k_v = 1132/30$), $R_v \approx 4.78 \times 10^5$.

- *Primary harmonic:* cycle-2 chain contribution at the chain-position assignment $k_v = 1132/30 \approx 37.733$. The denominator $30 = h$ is the Coxeter number; the numerator 1132 is a structural-integer assignment from the heterotic Wilson-line breaking pattern (Ref. [12]; explicit derivation from CY3 cohomology is outside the present scope).
- *Sub-harmonic correction:* $\alpha_v = (\text{rank} \cdot h/m_{\max})\alpha_\tau^2 = (8 \cdot 30/29)\alpha_\tau^2$. The coefficient $\text{rank} \cdot h/m_{\max}$ encodes the rank-locked Cartan-subspace projection at order α_τ^2 (paralleling the muon's order-2 sub-harmonic).
- *Sector modifier:* cycle-2 chain.
- *Dark-sector leakage:* absent for the vev itself, since the Higgs vev is the substrate baseline tension and does not couple to the spectrum's upper boundary as a fluctuation mode would.

b. Higgs boson, $R_H \approx 2.43 \times 10^5$.

- *Primary harmonic:* same $\sqrt{2}$-chain primary $(\sqrt{2})^{k_v}$ as the vev, divided by 2 (the breathing-mode amplitude is half of the substrate baseline tension; the Higgs boson is the fluctuation mode around the vev, not a separate Coxeter position).
- *Sub-harmonic correction:* $\alpha_H = ((\text{rank} - 2)h/m_{\max})\alpha_\tau^2 = (180/29)\alpha_\tau^2$. The $\text{rank} - 2 = 6$ coefficient drops two Cartan generators relative to the vev correction α_v, reflecting the two fewer rank-locked directions accessible to the breathing mode.
- *Sector modifier:* cycle-2 $\sqrt{2}$-chain.
- *Dark-sector leakage:* $(1 + \delta_H)$ with $\delta_H = 1/(h + m_{\max}) = 1/59$. The Higgs boson, as a scalar, couples isotropically to the spectrum's upper boundary at the boundary-capacity scale $h + m_{\max} = 59$, giving the leading dark-sector leakage with no internal-index suppression. The sign of δ_H is positive (the leakage *enhances* the boson's amplitude, in contrast to the rank-locked correction α_H which reduces it).

c. W boson ($k_W = 34.5$), $R_W \approx 1.56 \times 10^5$.

- *Primary harmonic:* cycle-2 $\sqrt{2}$-chain contribution $(\sqrt{2})^{34.5}$. The half-integer Coxeter position reflects the W's SU(2) vector index structure (rather than scalar or integer-spinor).
- *Sub-harmonic correction:* $\alpha_W = (2/e_2)\alpha_\tau^2/(1 + \alpha_\tau/h_{SU(3)})$, with leading coefficient $2/e_2 = 2/7$ (vector multiplicity 2 over the substrate's second Coxeter slot $e_2 = 7$, the same slot that appears in $\sin^2\theta_{23}^{\rm PMNS} = \text{rank}/(2e_2) = 4/7$ and $\sin^2\theta_{13}^{\rm PMNS} = 2/(e_2 e_4) = 2/91$), and closure denominator $h_{SU(3)} = 3$ (colour-projected closure shared with α_H and α_μ).
- *Sector modifier:* cycle-2 $\sqrt{2}$-chain at half-integer position, shared with the heavy-quark $\sqrt{2}$-chain structure.
- *Dark-sector leakage:* $(1 + \delta_W)$ with $\delta_W = \delta_H/e_6 = 1/(59 \cdot 19) = 1/1121$, where $e_6 = 19$ is the E_8 exponent palindromically paired with the muon's position $k_\mu = 11 = e_3$ (since $11 + 19 = h$). The factor of 19 suppression relative to δ_H encodes the W's internal SU(2) vector index structure projecting onto the $(11, 19)$-paired exponent. The sign of δ_W is positive, like δ_H.

d. Z boson and gauge couplings. The Z boson, g_2, g_1, and λ_H are derived bookkeeping quantities: $m_Z = m_W/\cos\theta_W$, $g_2 = 2m_W/v$, $g_1 = g_2 \tan\theta_W$, $\lambda_H = m_H^2/(2v^2)$. These are not independent structural predictions; they inherit precision from m_W, v, m_H, and $\sin^2\theta_W$ via standard SM tree-level relations.

VI. ELECTROWEAK GAUGE SECTOR

A. Weak mixing angle

The weak mixing angle takes the closed-form structural expression

$$\boxed{\sin^2\theta_W \;=\; \frac{2}{9}+\frac{\Delta_\varphi}{h} \;=\; \frac{2}{9}+\frac{1}{h(h+\varphi^2)},} \tag{36}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio and $\Delta_\varphi = 1/(h+\varphi^2)$. The leading rational term $2/9$ is, in this framework, the $SU(3)_{\text{color}}$-induced bulk projection at the unbroken-electroweak-symmetry locus (the geometric weight $2/(h_{SU(3)}\cdot h^{\text{extended}}_{SU(3)}) = 2/(3\cdot 3)$ assigned to the visible-sector $SU(2)_L\times U(1)_Y$ subprojection within the bulk symmetry breaking ansatz). We flag this as a *framework structural assignment* rather than a standard derived E_8 identity: it is the parsimonious choice that reproduces the observed value to within the small icosahedral correction below, and the assignment must ultimately be justified at the level of the underlying $E_8\times E_8$ heterotic compactification on $X/(\mathbb{Z}_3\times\mathbb{Z}_3)$. The small correction Δ_φ/h is the golden-ratio-conjugate subharmonic contribution from the visible E_8 embedding, with the same icosahedral provenance as the McKay-correspondence integer 5 in the gateway formula. Numerically, $\sin^2\theta_W = 0.222\overline{2} + 0.001022 = 0.22324$, agreeing with the on-shell PDG value $1 - m_W^2/m_Z^2 = 0.22320$ at 198 ppm.

B. Gauge couplings and Z boson

The $SU(2)_L$ and $U(1)_Y$ gauge couplings follow from the predicted m_W, v, and θ_W via the standard SM tree-level relations:

$$g_2 \;=\; \frac{2m_W}{v} \;=\; 0.65283, \tag{37}$$

$$g_1 \;=\; g_2\tan\theta_W \;=\; \frac{2m_W\tan\theta_W}{v} \;=\; 0.34998, \tag{38}$$

$$e \;=\; g_2\sin\theta_W, \qquad \alpha_{\text{em}} \;=\; \frac{e^2}{4\pi\epsilon_0\hbar c}. \tag{39}$$

The Z boson mass is

$$\begin{aligned} m_Z \;&=\; \frac{m_W}{\cos\theta_W} \\ &=\; \frac{h_{\text{eff}}\cdot k_W\cdot(1+\delta_W)\cdot K_{\text{cycle},W}}{\cos\theta_W} \\ &=\; 91.19037\ \text{GeV}, \end{aligned} \tag{40}$$

agreeing with PDG 2024 at 30 ppm.

a. Heterotic-string expectation: gauge unification. A heterotic $E_8\times E_8$ compactification on a Calabi–Yau threefold normally predicts that the running visible-sector gauge couplings $\alpha_1(\mu),\alpha_2(\mu),\alpha_3(\mu)$ unify at a GUT scale set by the compactification volume, $M_{\text{GUT}}\sim 10^{16}$ GeV in conventional units, with $\alpha_{\text{GUT}}\sim 1/24$. Within the present framework, g_1, g_2, and g_3 are predicted at the Z pole as separate structural objects rooted in distinct E_8 invariants (k_W, $\sin^2\theta_W$, α_s), and the framework does not predict their evolution to high scale: the reconciliation of the predicted low-scale couplings with one-loop unification of the running couplings is left for future work, contingent on a specific identification of the substrate-physical interpretation of RG running. We note that the Z-pole prediction $\alpha_s(M_Z) = 0.1180$ is consistent at the 5% level with unification at M_{GUT} given two-loop SM running and threshold corrections at $M_{\text{SUSY}}\sim 1$ TeV; no claim beyond consistency is made.

C. Higgs self-coupling

The Higgs self-coupling is the standard SM tree-level relation inverted to read off λ_H:

$$\begin{aligned} \lambda_H \;&=\; \frac{m_H^2}{2v^2} \\ &=\; \frac{[h_{\text{eff}}(k_v-2)(1-\alpha_H)(1+\delta_H)]^2}{2[h_{\text{eff}}k_v(1-\alpha_v)]^2} \;=\; 0.12928011, \end{aligned} \tag{41}$$

agreeing with the observed $\lambda_H = 0.12928057$ at 3.5 ppm.

D. Strong coupling

The strong coupling at the Z pole takes the closed-form structural expression

$$\boxed{\begin{aligned} \alpha_s(M_Z) \;&=\; \frac{m_{\max}^2+\text{rank}}{m_{\max}\cdot\text{dim}} \;=\; \frac{29^2+8}{29\cdot 248} \\ &=\; \frac{849}{7192} \;=\; 0.118048, \end{aligned}} \tag{42}$$

which uses only forced E_8 algebra invariants $\{\text{dim},\text{rank},m_{\max}\}$. The numerator $m_{\max}^2+\text{rank}$ combines the squared largest exponent $m_{\max}^2 = (h-1)^2 = 841$ with the algebra rank rank $= 8$; the denominator $m_{\max}\cdot\text{dim} = 29\cdot 248 = 7192$ is the largest-exponent times the algebra dimension. *We flag this as a framework structural assignment.* Although the numerator and denominator are each built from E_8 invariants, neither $m_{\max}^2+\text{rank}$ nor $m_{\max}\cdot\text{dim}$ corresponds directly to a standard Lie-algebraic invariant (quadratic Casimir, Killing form, dimension of an irrep, etc.): the quadratic Casimir of the adjoint of E_8 is $C_2(\text{adj}) = 2h^\vee = 60$ (with dual Coxeter number $h^\vee = h$ for simply-laced groups), and no standard identity recovers 849/7192 from it. The construction selects the unique combination of $\{\text{dim},\text{rank},m_{\max}\}$ at a fixed weight that reproduces the PDG value to within experimental precision; a first-principles identification

(e.g. via a specific gauge-boundary-condition trace on $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$) is left for future work. Numerically $\alpha_s(M_Z) = 0.118048$, agreeing with the PDG 2024 world average $\alpha_s(M_Z) = 0.1180 \pm 0.0009$ at 405 ppm, well within experimental precision. This is the only closed-form structural expression for $\alpha_s(M_Z)$ in the present framework; no continuous parameter enters.

E. Fine-structure constant

The fine-structure constant follows from g_2 and θ_W via $e = g_2 \sin\theta_W$ and $\alpha_{\rm em} = e^2/(4\pi)$ in natural units. With Eqs. (37) and (36),

$$\alpha_{\rm em} = \frac{(g_2 \sin\theta_W)^2}{4\pi} = \frac{(2m_W \sin\theta_W/v)^2}{4\pi}, \tag{43}$$

inheriting precision from m_W, v, and $\sin\theta_W$. The on-shell predicted value matches the PDG within experimental precision (Sec. XIII).

VII. STRONG CP ANGLE

The $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient structure provides a geometric route to a vanishing QCD vacuum angle, but the required cancellation is not derived independently here. The QCD topological term integrates as

$$\int_{X/(\mathbb{Z}_3\times\mathbb{Z}_3)} \mathrm{Tr}(F \wedge F) = \frac{1}{|\mathbb{Z}_3 \times \mathbb{Z}_3|} \int_X \mathrm{Tr}_g(F \wedge F), \tag{44}$$

where Tr_g is twisted by the group action. For the canonical E_8 embedding and Wilson-line data [12, 13], the present framework assumes that the quotient projection annihilates the relevant twisted trace. The mathematical content of this assumption is the equivariant-cohomology identity

$$\big[\mathrm{ch}_2(V) \cdot [g]\big]_X = 0 \quad \text{for every nontrivial } g \in \mathbb{Z}_3 \times \mathbb{Z}_3, \tag{45}$$

where $\mathrm{ch}_2(V) = \frac{1}{2}\,\mathrm{Tr}(F \wedge F)/(2\pi)^2$ is the second Chern character of the gauge bundle V on the cover X and $[g]$ is the cohomology class of the fixed locus of g (empty for a freely-acting $\mathbb{Z}_3 \times \mathbb{Z}_3$ on the Schoen threefold, which is the case of interest here). Equivalently, the assumption is that the descent of $\mathrm{Tr}(F \wedge F)$ to $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ admits no non-zero g-twisted contribution under the orbifold-cohomology decomposition. Under that assumption, $\mathrm{Tr}_{\mathbb{Z}_3\times\mathbb{Z}_3}\mathrm{Tr}(F \wedge F) = 0$, which would set $\theta_{\rm QCD} = 0$. The heterotic construction papers fix the quotient geometry and bundle data on which this statement depends, but they do not themselves prove Eq. (45) for the specific bundle data the framework needs. Within the framework, this therefore functions as a geometric strong-CP mechanism rather than as a separately established theorem. If the cancellation holds, the strong CP problem is removed without Peccei–Quinn [30] or a dynamical axion [29].

VIII. NEUTRINO SECTOR

A. Mass eigenvalues from sub-gateway φ-chain modes

The neutrino mass eigenvalues arise as deeply sub-gateway φ-chain modes at recursive depths $n = 3, 4$ in α_τ. Their explicit closed-form ratios are

$$R_{\nu_1} = 2\,\varphi^2\,\alpha_\tau^4, \tag{46}$$

$$R_{\nu_3} = 2\,\alpha_\tau^3\,\big(1 + 5\varphi^2\alpha_\tau\big), \tag{47}$$

$$R_{\nu_2} = R_{\nu_3} \cdot \frac{1+\Delta_\varphi}{\mathrm{rank}-2} = R_{\nu_3} \cdot \frac{1+\Delta_\varphi}{6}, \tag{48}$$

where ν_1 is the lightest mass eigenstate (4th-order icosahedral mode with φ^2 prefactor), ν_3 is the heaviest (3rd-order with sub-leading icosahedral correction $5\varphi^2\alpha_\tau$), and ν_2 relates to ν_3 by the icosahedral-bulk closure $(1+\Delta_\varphi)$ divided by $\mathrm{rank} - 2 = 6$ (the number of Cartan directions accessible to the second-generation neutrino mode). Numerically, $m_{\nu_1} \approx 0.451$ meV, $m_{\nu_2} \approx 8.61$ meV, $m_{\nu_3} \approx 50.21$ meV.

B. Mass-squared splittings and bounds

The mass-squared splittings derived from Eqs. (46)–(48): $\Delta m^2_{21} = 7.4193 \times 10^{-5}$ eV^2 (90 ppm), $\Delta m^2_{31} = 2.5211 \times 10^{-3}$ eV^2 (2440 ppm), agreeing with the NuFIT 5.3 global-fit ranges [34, 35] within experimental precision.

Normal mass ordering is selected with $\Sigma m_\nu = 59.3$ meV, KATRIN effective mass $m_\beta = 8.82$ meV, and $m_{\beta\beta} = 4.00$ meV (assuming Majorana nature with vanishing Majorana phases). These remain below current cosmological, direct-kinematic, and neutrinoless-double-beta-decay bounds from DESI, KATRIN, and KamLAND-Zen [36–38].

a. Dirac vs Majorana nature. The framework's mass-eigenvalue construction $(R_{\nu_1}, R_{\nu_2}, R_{\nu_3})$ does not by itself distinguish Dirac from Majorana neutrinos: the closed-form ratios specify the three positive eigenvalues of $|m_\nu|$, not the Lorentz-structure of the mass term. The framework's heterotic $E_8 \times E_8$ embedding on $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ admits both possibilities, with right-handed neutrinos arising as either standalone Dirac partners or as integrated-out heavy seesaw modes producing an effective Majorana mass at low scale. The $m_{\beta\beta} = 4.00$ meV prediction quoted above is the Majorana-case value with vanishing Majorana phases; the corresponding $m_{\beta\beta}$ in the Dirac case is identically zero. A future framework-internal derivation determining which case obtains would convert $m_{\beta\beta}$ from a conditional prediction into a direct falsifiable observable; the present construction commits to the mass-eigenvalue triplet only.

C. PMNS mixing angles and CP phase

The PMNS lepton-mixing parameters take explicit closed forms involving E_8 exponents and the $SU(3)_{\rm color}$ Coxeter number $h_{SU(3)} = 3$:

$$\sin^2\theta_{12}^{\rm PMNS} = \frac{h}{h_{SU(3)}}\cdot\Delta_\varphi = \frac{10}{h+\varphi^2}, \tag{49}$$

$$\sin^2\theta_{23}^{\rm PMNS} = \frac{\rm rank}{2\,e_2} = \frac{4}{7}, \tag{50}$$

$$\sin^2\theta_{13}^{\rm PMNS} = \frac{2}{e_2\cdot e_4} = \frac{2}{91}, \tag{51}$$

$$\delta_{\rm CP} = \frac{e_3\cdot h_{SU(3)}\cdot\pi}{h} = \frac{11\cdot 3\cdot\pi}{30} = \frac{11\pi}{10}, \tag{52}$$

where $e_2 = 7, e_3 = 11, e_4 = 13$ are the second, third, and fourth E_8 Coxeter exponents respectively. The CP phase $\delta_{\rm CP} = 11\pi/10 = 198^\circ$ has numerator equal to the muon's Coxeter step $k_\mu = 11$ $(= e_3)$ times $h_{SU(3)} = 3$, denominator $h = 30$. All four PMNS predictions agree with Nu-FIT 5.3 within experimental precision. The small-integer combinations of $\{\text{rank}, e_j, h, h_{SU(3)}, \Delta_\varphi\}$ chosen for each PMNS angle are constrained but not uniquely forced by E_8 algebra alone; the framework's structural choices for these combinations are dictated by the requirement that the closed forms match observation, and a first-principles derivation from the underlying CY3 cohomology is left for future work.

IX. QUARK MIXING (CKM)

The CKM matrix elements take explicit closed forms using the same structural integers:

$$\sin^2\theta_{12}^{\rm CKM} = \frac{h/2-1}{\dim+h} = \frac{14}{278}, \tag{53}$$

$$\sin^2\theta_{23}^{\rm CKM} = \frac{4\sin^2\theta_{12}^{\rm CKM}}{e_3^2}\cdot\Big(1+\frac{\Delta_\varphi}{h_{SU(3)}}\Big), \tag{54}$$

$$\sin^2\theta_{13}^{\rm CKM} = \alpha_\tau^2\cdot\big(1+4\Delta_\varphi\big), \tag{55}$$

$$\delta_{\rm CKM} = \frac{e_3\cdot\pi}{h} = \frac{11\pi}{30} = 66.0^\circ. \tag{56}$$

The Cabibbo angle uses the chain depth $h/2-1 = M = 14$; the atmospheric mixing involves the muon-exponent $e_3 = 11$ squared in the denominator with an icosahedral-bulk correction $\Delta_\varphi/h_{SU(3)}$; the reactor mixing $\sin^2\theta_{13}^{\rm CKM} = \alpha_\tau^2(1+4\Delta_\varphi)$ is at order α_τ^2 with a four-fold icosahedral correction. All four CKM predictions agree with PDG 2024 within experimental precision. The leptonic CP phase $\delta_{\rm CP} = 11\pi/10$ and the quark CP phase $\delta_{\rm CKM} = 11\pi/30$ share the same E_8-exponent numerator 11, differing only by the factor of $h_{SU(3)} = 3$ that distinguishes the lepton sector from the quark sector. As with the PMNS angles, the specific small-integer combinations chosen for each CKM angle (e.g., the factor of 4 in Eq. (54), the factor of 4 in front of Δ_φ in Eq. (55)) are constrained but not uniquely forced by E_8 algebra; their first-principles derivation from the heterotic CY3 cohomology is left for future work.

X. DARK-MATTER TOWER

A. The unified neutrino-DM Coxeter tower

The framework's Coxeter chain produces a unified neutrino-and-dark-matter tower:

$$\boxed{m_n = 2\,h_{\rm eff}\,\alpha_\tau^n,} \tag{57}$$

with $\alpha_\tau = 1/(\dim+h) = 1/278 \approx 3.5971\times10^{-3}$ and the icosahedral spin-doublet prefactor $c_n = 2$ shared with ν_3. The electron-anchor value of $h_{\rm eff}$ is 0.5151534130081301 MeV (the muon-anchor differs by 0.0011 ppm).

Each order n defines a structurally distinct mode with mass m_n, characteristic kinetic-mixing parameter $\epsilon_n \equiv \alpha_\tau^n$, and de Broglie wavelength $\lambda_{\rm dB,n} = h/(m_n v_{\rm DM})$ at the galactic virial velocity $v_{\rm DM} = 220$ km/s $= 7.3384\times10^{-4}\,c$.

B. Cosmological cutoffs and viable mass window

Two cosmological cutoffs bracket the viable dark-matter window:

- **Lyman-α cutoff**: ultralight bosonic dark-matter modes with $m \lesssim 2\times10^{-20}$ eV are excluded at 95% C.L. by Lyman-α forest constraints [40].
- **Hubble coherence cutoff**: modes lighter than $H_0 \sim 1.4\times10^{-33}$ eV cannot oscillate coherently in the matter-dominated era and are believed to behave as dark energy rather than dark matter; the precise role of substrate modes at orders $n \geq 13$ in the present framework, including any contribution to the cosmological constant, will require further research.

These cutoffs partition the tower (Eq. (57)) into three regimes: visible-sector neutrinos at orders $n = 3, 4$; dark matter at orders $n = 5$–11 (seven discrete modes); and excluded modes at $n \geq 12$. Table X lists every order from $n = 3$ to $n = 15$ with its mass, kinetic-mixing parameter, de Broglie wavelength at $v_{\rm DM}$, and physical regime.

C. Order-5 dominant mode: couplings and detection prospects

The dominant dark-matter mode is at $n = 5$:

$$m_5 = 2\,h_{\rm eff}\,\alpha_\tau^5 = 620.5012 \text{ neV}, \tag{58}$$

TABLE X. Unified neutrino–dark-matter Coxeter tower from order $n = 3$ (heaviest visible neutrino) through order $n = 15$ (super-Hubble). Mass: $m_n = 2h_{\rm eff}\alpha_\tau^n$ with $\alpha_\tau = 1/(\dim + h) = 1/278$. Kinetic-mixing parameter: $\epsilon_n = \alpha_\tau^n$. de Broglie wavelength at galactic virial velocity $v_{\rm DM} = 220$ km/s.

n	Prefactor	Mass	$\lambda_{\rm dB}$	ϵ_n	Regime
3	$2\,\alpha_\tau^3$	47.9548 meV	5.6073 mm	$4.654e-08$	(nu_3, nu_2, visible)
4	$2\,\alpha_\tau^4$	172.4993 ueV	1.5588 m	$1.674e-10$	(nu_1, visible)
5	$2\,\alpha_\tau^5$	620.5012 neV	433.3531 m	$6.022e-13$	ALP / dark-photon (ADMX)
6	$2\,\alpha_\tau^6$	2.2320 neV	120.4722 km	$2.166e-15$	ALP / atomic-clock DM
7	$2\,\alpha_\tau^7$	8.0288 peV	33491.2600 km	$7.793e-18$	atomic-clock DM
8	$2\,\alpha_\tau^8$	28.8807 feV	9310570.2763 km	$2.803e-20$	atomic-clock / pulsar-timing
9	$2\,\alpha_\tau^9$	103.8875 aeV	17.3017 AU	$1.008e-22$	pulsar-timing / asteroid
10	$2\,\alpha_\tau^{10}$	373.6961 zeV	4809.8804 AU	$3.627e-25$	wave-DM
11	$2\,\alpha_\tau^{11}$	1.3442 zeV	6.4821 pc	$1.305e-27$	wave-DM, sub-galactic
12	$2\,\alpha_\tau^{12}$	4.8354 yeV	1.8020 kpc	$4.693e-30$	EXCLUDED by Lyman-alpha
13	$2\,\alpha_\tau^{13}$	17.3934 rdeV	500.9615 kpc	$1.688e-32$	EXCLUDED (super-Hubble)
14	$2\,\alpha_\tau^{14}$	0.0626 rdeV	139.2673 Mpc	$6.073e-35$	EXCLUDED (super-Hubble)
15	$2\,\alpha_\tau^{15}$	0.0002 rdeV	38716.3076 Mpc	$2.184e-37$	EXCLUDED (super-Hubble)

with structurally-fixed kinetic mixing

$$\epsilon_5 = \alpha_\tau^5 = 6.022 \times 10^{-13} \tag{59}$$

and axion-photon coupling derived from ϵ_5, $\alpha_{\rm em}$, and the predicted Higgs vev:

$$g_{a\gamma\gamma} = \frac{\epsilon_5\,\alpha_{\rm em}}{v} = 1.7849 \times 10^{-17}\ {\rm GeV}^{-1}. \tag{60}$$

The mass falls in the upper end of the ADMX direct-detection sensitivity window for axion-like and dark-photon dark-matter searches.

D. Geometric-series abundance convergence

The same factor $\alpha_\tau = 1/278$ that produces successive mass levels also fixes the relative cosmological abundance. The order-5 contribution is dominant; orders 6–11 contribute via the convergent geometric series:

$$\zeta_{\rm tower} \equiv \sum_{n=5}^{11} \frac{\Omega_n}{\Omega_5} = \sum_{k=0}^{6} \alpha_\tau^k \approx 1.0638. \tag{61}$$

The convergence is rapid (Table XI); orders 6–11 contribute only $\approx 6.0\%$ of the total dark-matter abundance, and order 5 contributes $\approx 94.0\%$.

E. Decay constant and misalignment angle

The order-5 decay constant and misalignment angle are fixed by structural integers and the predicted electroweak / Planck scales:

TABLE XI. Geometric-series abundance convergence over the viable dark-matter window $n = 5$–11. The cumulative ratio saturates at $\zeta_{\rm tower} \approx 1.0638$.

n	Ω_n/Ω_5	Cumulative $\sum_{n'=5}^{n} \Omega_{n'}/\Omega_5$
5	1	1.000000
6	5.998×10^{-2}	1.059976
7	3.597×10^{-3}	1.063573
8	2.157×10^{-4}	1.063789
9	1.294×10^{-5}	1.063802
10	7.760×10^{-7}	1.063803
11	4.654×10^{-8}	1.063803

$$\begin{aligned} f_5 &= \frac{\mathrm{rank}(h-\mathrm{rank})}{\dim} \cdot \frac{\sqrt{v \cdot m_{\rm Planck}}}{\alpha_\tau} \\ &= \frac{22}{31} \cdot \frac{\sqrt{v\, m_{\rm Planck}}}{\alpha_\tau} \\ &= 1.082 \times 10^{13}\ {\rm GeV}, \end{aligned} \tag{62}$$

$$\theta_5 = \arctan\varphi \approx 1.0172\ {\rm rad} \approx 58.28^\circ. \tag{63}$$

The ratio $\mathrm{rank}(h-\mathrm{rank})/\dim = 8 \cdot 22/248 = 22/31$ is the off-diagonal Coxeter coupling prefactor. The $\sqrt{v \cdot m_{\rm Planck}}$ factor is the geometric mean of the electroweak and Planck scales (the "substrate see-saw amplitude"). The misalignment angle $\theta_5 = \arctan\varphi$ uses the icosahedral golden ratio.

F. Rank-locked sub-harmonic correction

A structural pattern of rank-locked-subspace corrections operates across the gateway-mixing tower (Table XII). The order-5 mode receives the rank-locked cor-

rection

$$m_5^{\text{corrected}} = 2h_{\text{eff}}\alpha_\tau^5 \cdot (1 - \text{rank}\,\alpha_\tau) = 2h_{\text{eff}}\alpha_\tau^5 \cdot \frac{270}{278} = 602.645 \text{ neV}, \quad (64)$$

a -2.878% shift from the bare tower value $m_5^{\text{bare}} = 620.501$ neV. The correction follows the same pattern of small-coefficient Coxeter sub-harmonic shifts that operate on τ, μ, v, and ν_3 (Table XII).

TABLE XII. Rank-locked sub-harmonic correction pattern across the gateway-mixing tower. Each correction takes the form $(1 \pm c_n\alpha_\tau^k)$ where the coefficient c_n and power k are fixed by the mode's structural assignment.

Mode	n	Correction	Coefficient
τ	1	$(1-\alpha_\tau)$	1
μ	2	$(1-2\alpha_\tau^2)$	2
v	2	$(1-\text{rank}\,\alpha_\tau^2)$	8
ν_3	3	$(1+5\varphi^2\alpha_\tau)$	$5\varphi^2 \approx 13.09$
m_5	5	$(1-\text{rank}\,\alpha_\tau)$	8

G. Term-by-term decomposition: dark-matter tower modes

The dark-matter tower mass formula $m_n = 2h_{\text{eff}}\alpha_\tau^n$ (Eq. (57)) admits the same four-component decomposition. Each order n in the tower is a sub-gateway recursive mode (analogous to the leptons but at deeper recursive depth α_τ^n rather than $h^{k/15}$).

a. Generic tower mode ($n \geq 3$).

$$m_n = \underbrace{2}_{\text{prefactor}} \cdot h_{\text{eff}} \cdot \underbrace{\alpha_\tau^n}_{\text{primary harmonic at depth } n} \cdot \underbrace{(1+\mathcal{C}_n)}_{\text{order-specific correction}}$$

- *Icosahedral spin-doublet prefactor:* the constant 2 is the icosahedral spin-doublet multiplicity, shared across all tower modes (including ν_3). Structurally, it encodes the binary icosahedral group's two-element kernel under the McKay correspondence.
- *Primary harmonic:* α_τ^n where $\alpha_\tau = 1/(\dim + h) = 1/278$ is the gateway-mixing parameter. Successive orders n scale by α_τ, generating a discrete geometric tower.
- *Sector modifier:* sub-gateway recursive (α_τ chain), distinct from the visible-sector φ-chain (leptons) and $\sqrt{2}$-chain (quarks).
- *Order-specific correction $\mathcal{C}_n$:* sub-harmonic shift specific to order n. For $n = 3$ (ν_3), $\mathcal{C}_3 = 5\varphi^2\alpha_\tau$ (icosahedral correction). For $n = 5$ (dominant DM), $\mathcal{C}_5 = -\text{rank}\,\alpha_\tau$ (rank-locked correction; see below). For $n = 4, 6, 7, \ldots, 11$, the corrections are sub-leading and do not change the tower's geometric-series structure.

b. Dominant DM mode ($n = 5$).

$$m_5 = 2 \cdot h_{\text{eff}} \cdot \underbrace{\alpha_\tau^5}_{\text{primary harmonic}} \cdot \underbrace{(1 - \text{rank}\,\alpha_\tau)}_{\text{rank-locked sub-harmonic}}$$

- *Primary harmonic:* $\alpha_\tau^5 = (1/278)^5 \approx 6.0 \times 10^{-13}$, the fifth recursive depth. Combined with $2h_{\text{eff}}$, this gives the bare value 620.5012 neV.
- *Sub-harmonic correction:* $(1 - \text{rank}\,\alpha_\tau) = (1 - 8/278) = 270/278 \approx 0.9712$, the rank-locked-subspace projection. The factor rank = 8 here is the same rank that appears at order α_τ^2 for the Higgs vev correction α_v, but here at first order in α_τ rather than second, reflecting the deeper recursive depth $n = 5$.
- *Sector modifier:* sub-gateway α_τ-chain.
- *Dark-sector leakage:* not separately applicable; the sub-gateway recursive structure is itself the substrate-internal coupling channel.

c. Couplings. The kinetic-mixing parameter $\epsilon_n = \alpha_\tau^n$ at each order is identified directly with the mixing parameter in the dark-photon literature; for $n = 5$, $\epsilon_5 = 6.022 \times 10^{-13}$. The axion-photon coupling $g_{a\gamma\gamma} = \epsilon_5\alpha_{\text{em}}/v$ follows from the substrate cross-term construction (developed in the companion paper [43]) extended to the dark-sector mode.

d. Structural pattern across the gateway-mixing tower. The rank-locked sub-harmonic corrections (Table XII) follow a discrete, mode-specific pattern with coefficients $\{1, 2, \text{rank}, 5\varphi^2, \text{rank}\}$ for the modes $\{\tau, \mu, v, \nu_3, m_5\}$. The repetition of rank = 8 in both α_v (vev, order α_τ^2) and m_5 (DM, order α_τ^1) reflects the rank-locked-subspace projection shared between the cycle-2 boson sector and the deepest sub-gateway DM mode.

H. Cosmological abundance result

Combining the corrected m_5 from Eq. (64), the structurally-derived f_5 from Eq. (62), the icosahedral misalignment angle θ_5 from Eq. (63), and the geometric-series tower-sum factor $\zeta_{\text{tower}} \approx 1.0638$, the standard misalignment-style abundance law [29], translated into the substrate variables used here, gives the consistency estimate

$$\boxed{\Omega_{\text{DM}}h^2 = 0.119982,} \quad (65)$$

consistent with the Planck 2018 measured value $\Omega_{\rm DM}h^2 = 0.1200 \pm 0.001$ [39] at 0.015 %, without introducing any additional fitted cosmological parameter. The inputs $\{f_5, \theta_5, m_5, \zeta_{\rm tower}\}$ are all derived from $E_8 \times E_8$ structure plus the same single electron-mass anchor that produces the Standard-Model spectrum, so this match represents a structural consistency check rather than an independent first-principles cosmological derivation.

We note that the imported misalignment-style abundance formula carries $O(1)$ order-of-magnitude uncertainty in its application across mass regimes; the precise numerical agreement at 0.015 % should be interpreted within that uncertainty as "order-of-magnitude consistent with Planck." A full substrate-native cosmological evolution computation (Klein–Gordon evolution through radiation–matter equality at m_5) is left for follow-up work; only after that step would the abundance value qualify as a first-principles prediction in the same sense as the mass-spectrum relations.

XI. GRAVITATIONAL SECTOR

Newton's constant from Eq. (6):

$$G^{\rm pred} = 6.67431691 \times 10^{-11}\ {\rm m^3 kg^{-1} s^{-2}}, \qquad (66)$$

agreeing with CODATA 2022 [32] at 2.5 ppm, below the G-measurement scatter that still characterizes direct laboratory determinations [33]. Planck mass: $m^{\rm pred}_{\rm Planck} = 1.22088858 \times 10^{19}$ GeV (1.2 ppm).

XII. TWO-ANCHOR DIAGNOSTIC CONSISTENCY CHECK

The framework's predictive path computes $h_{\rm eff} = v_C/R_v$ from the structural Higgs-vev identity (Sec. I); no PDG mass enters. As a diagnostic check on the framework's internal arithmetic, we additionally evaluate two fermion-anchored modes (anchoring $h_{\rm eff}$ on the CODATA electron mass or on the CODATA muon mass) and compare their predictions across the entire 26-parameter spectrum. The structural-vev mode is the primary prediction; the two fermion-anchored modes are diagnostic only. Cross-checking the two fermion-anchored modes against each other gives a single consistency residual

$$\Delta_{e\mu} = \frac{F_\mu/F_e}{m_\mu/m_e} - 1 = -2.98 \times 10^{-8} = -30 \text{ ppb}, \qquad (67)$$

constant across all 26 SM observables. The residual is set by sub-harmonic-series truncation, not structural ambiguity. For derived quantities involving $1/m^2_{\rm Planck}$, the residual doubles: G shows two-anchor variance $2|\Delta_{e\mu}| =$ 60 ppb, observed.

XIII. RESULTS

a. Scheme and renormalization conventions. The structural mass predictions in this paper are matched against PDG values in the schemes in which those values are reported, and *the schemes mix across rows.* We adopt the following explicit conventions, row by row, in Table XIII.

- *Charged leptons* (m_e, m_μ, m_τ): pole/physical masses (the QED self-energy is infrared-finite for charged leptons so this coincides with the direct experimental measurement).
- *Light quarks* (m_u, m_d, m_s): $\overline{\rm MS}$ at $\mu = 2$ GeV.
- *Heavy quarks* (m_c, m_b): $\overline{\rm MS}$ at $\mu = m_q$ (i.e. $\overline{\rm MS}(m_c)$ and $\overline{\rm MS}(m_b)$).
- *Top quark* (m_t): the world-average direct (kinematic) mass quoted by PDG, which is closest to the *pole-mass* convention, *not* $\overline{\rm MS}(m_t)$. The latter is ~ 163 GeV, well below the framework's prediction ~ 172.8 GeV; the framework's match is to the direct/pole value 172.57 GeV.
- *Gauge bosons* (m_W, m_Z, m_H): pole masses.
- *Higgs vev* (v): tree-level minimum of the Higgs potential at the Z-pole scale.
- *Weak mixing angle*: on-shell scheme, $\sin^2\theta_W^{\rm OS} \equiv 1 - m_W^2/m_Z^2$.
- *Strong coupling*: $\alpha_s^{\overline{\rm MS}}(\mu = M_Z)$.
- *Fine-structure constant*: Thomson-limit $\alpha_{\rm em}(0) = 1/137.036$.

Mixed-scheme caveat. Because the framework's predictions match against pole/physical masses for some particles (leptons, top, gauge bosons, Higgs), $\overline{\rm MS}$ at fixed external scale for others (light and heavy quarks), the on-shell scheme for $\sin^2\theta_W$, the $\overline{\rm MS}(M_Z)$ scheme for α_s, and the Thomson-limit value for $\alpha_{\rm em}$, the ppm-level agreements reported in Table XIII should be read as agreement with PDG values *within each row's stated scheme* and *not* as agreement of a single uniformly-defined object across rows. In particular, switching m_t to $\overline{\rm MS}(m_t)$, $\sin^2\theta_W$ to $\overline{\rm MS}(M_Z)$, or $\alpha_{\rm em}$ to $\alpha_{\rm em}(M_Z)$ would each shift the comparison by amounts $O(1$–$10\%)$, far above the ppm headline. The framework's structural sub-harmonic corrections ($\alpha_\tau, \alpha_v, \alpha_H, \delta_H, \delta_W$, the per-quark χ_q, λ_q signs, etc.) are identified with the all-orders effective corrections in each row's scheme: they absorb what would otherwise be separately computed perturbative loop corrections in a Lagrangian derivation. The framework therefore predicts only the final scheme-specified value, and the schemes are different for different rows by physics convention rather than by choice. A first-principles Lagrangian derivation in which the structural corrections

are matched against explicit one- and two-loop calculations in a *single* unified scheme is left for future work; that derivation would convert mixed-scheme ppm-level agreement into single-scheme ppm-level agreement.

b. Falsification implication. Future PDG updates that move any single row's observable outside the framework's prediction at the ppm level *in that row's own stated scheme* constitute direct falsification. Cross-row re-expression in a unified scheme is not committed to and would require the future Lagrangian-level derivation noted above.

Table XIII summarizes the full prediction set. Primary observables are listed together with four algebraically derived consistency checks that are not counted as independent experimental tests.

c. Probe-scale dependence and the deviation pattern. The deviation column of Table XIII is not flat; it exhibits a structured pattern that tracks the experimental *probe scale* Q at which each PDG value is extracted, and the associated theoretical uncertainty in that extraction. Reading the table from cleanest to noisiest:

- *Charged leptons* ($Q \sim m_\ell$, pure-QED extraction). m_e is the framework anchor (zero by construction); m_μ agrees at 1.1 ppb (the precision of muonium spectroscopy); m_τ at 0.15 ppm (the precision of τ-pair production at e^+e^- thresholds). PDG-quoted experimental uncertainties on these masses are themselves at $\lesssim 10^{-7}, 10^{-7}, 10^{-4}$ relative respectively, and the framework's deviations track them.

- *Z pole electroweak observables* ($Q \sim M_Z$, on-shell or $\overline{\mathrm{MS}}$ extractions). m_Z agrees at 30 ppm, m_W at 4 ppm, $\sin^2\theta_W^{\mathrm{OS}}$ at 198 ppm, $\alpha_s^{\overline{\mathrm{MS}}}(M_Z)$ at 405 ppm. PDG experimental relative uncertainties at this scale are $\sim 10^{-5}, 10^{-4}, 10^{-3}, 10^{-3}$ respectively; again the framework's deviations track the experimental error budgets.

- *Heavy quarks* ($Q \sim m_q$, $\overline{\mathrm{MS}}(m_q)$ or pole/direct extraction). m_c at 1019 ppm, m_b at 974 ppm, m_t at 1293 ppm, all within the $\sim 10^{-3}$ relative PDG uncertainty band that itself reflects lattice-QCD systematics, scheme-conversion ambiguities (especially the renormalon-induced pole-vs-$\overline{\mathrm{MS}}$ split for the top), and Coulombic-bound state corrections at the b and c scales.

- *Light quarks* ($Q \sim 2$ GeV, $\overline{\mathrm{MS}}(2\text{ GeV})$ from chiral perturbation theory plus lattice). m_u at 5618 ppm, m_d at 755 ppm, m_s at 1577 ppm. Here the PDG-quoted uncertainties are $\sim 10^{-2}$ relative for m_u, m_d (FLAG-averaged lattice results) and $\sim 5 \times 10^{-3}$ for m_s. The framework's $\sim 0.5\%$–0.6% deviations on m_u, m_s are within this band; the 0.08% on m_d is well within. The 0.6% on m_u in particular is comparable to FLAG's own $\overline{\mathrm{MS}}(2\text{ GeV})$ chiral-extrapolation systematic.

- *Mixing angles and CKM/PMNS phases* (Q varying with extraction; PDG uncertainties percent-level). All within the percent-level PDG uncertainties.

- *Cosmological / gravitational sector* ($\Omega_{\mathrm{DM}}h^2$ from CMB at horizon scales; G at torsion-balance laboratory scales). Agreement at 152 ppm and 2.5 ppm respectively, within experimental error.

The framework's deviations therefore correlate with the experimental probe scale and extraction precision rather than with any framework-internal hierarchy: cleaner extractions yield sharper agreement, and noisier extractions yield deviations within their own uncertainty bands. We do *not* claim that this correlation alone establishes the framework — the residual deviations could in principle be hiding a systematic offset of a single multiplicative scale that happens to be inside all PDG error bars — but it does establish that the framework is not *tuned* to one probe scale at the expense of others. Future PDG updates that shrink the experimental uncertainty in any row will tighten the falsification window in that row.

XIV. PREDICTIONS AND FALSIFICATION

The framework makes pre-registered predictions across multiple sectors, falsifiable by:

1. Inverted neutrino mass ordering in future oscillation data.

2. Cosmological convergence to $\Sigma m_\nu > 70$ meV in future CMB and large-scale-structure analyses.

3. δ_{CP} outside $198^\circ \pm 5^\circ$ in future long-baseline oscillation data.

4. $0\nu\beta\beta$ at effective Majorana mass significantly above 4 meV (assuming Majorana nature) in next-generation neutrinoless-double-beta-decay searches.

5. Neutron EDM above the current upper bound, or a confirmed nonzero value above $\sim 10^{-30}\, e \cdot \mathrm{cm}$ in next-generation EDM searches [31].

6. Newton's G convergence inconsistent with $6.67431691 \times 10^{-11}$ at ppm precision.

7. Dark-matter detection at masses inconsistent with the predicted tower (orders 5–11), or coupling significantly different from $g_{a\gamma\gamma} = 1.78 \times 10^{-17}\ \mathrm{GeV}^{-1}$.

8. Precision EW measurements converging to $\sin^2\theta_W$ inconsistent with $2/9 + \Delta_\varphi/h$ at ppm precision.

TABLE XIII. Complete table (Part 1 of 2): substrate-framework predictions for the SM mass spectrum, gauge couplings, mixing parameters, and strong-CP angle, anchored on the extended structural identity for the Higgs vev $v_C = (\dim -2)/(1-\delta_W)\cdot(1+5\alpha_\tau^3/8)$. Deviations are reported in ppm (parts per million). Continued in Table XIV.

Observable	Unit	Observed	Predicted	Deviation (ppm)
CHARGED LEPTON MASSES				
electron	MeV	0.5109989500	0.5109989501	+0.0002
muon	MeV	105.6583755000	105.6583756375	+0.0013
tau	MeV	1776.8600000000	1776.8597391214	-0.1468
QUARK MASSES				
up	MeV	2.1600000000	2.1721350187	+5618.0642
down	MeV	4.6700000000	4.6664726684	-755.3173
strange	MeV	93.4000000000	93.2527287672	-1576.7798
charm	MeV	1273.0000000000	1274.2976607964	+1019.3722
bottom	MeV	4183.0000000000	4178.9244869807	-974.3039
top	GeV	172.5700000000	172.5698833015	-0.6762
NEUTRINO MASSES				
neutrino 1	meV	0.4510000000	0.4516090892	+1350.5303
neutrino 2	meV	8.6139000000	8.6253795301	+1332.6751
neutrino 3	meV	50.1498000000	50.2128570823	+1257.3746
ELECTROWEAK SECTOR				
Higgs vev	GeV	246.2196500000	246.2196500197	+0.0001
Higgs boson	GeV	125.2000000000	125.1997813542	-1.7464
W boson	GeV	80.3692000000	80.3692003835	+0.0048
Z boson	GeV	91.1876000000	91.1900348225	+26.7012
Weinberg mixing fraction		0.223200000000	0.223244151789	+197.8127
SU(2) coupling		0.653300000000	0.652825234518	-726.7189
U(1)_Y coupling (EW norm)		0.350000000000	0.349981197590	-53.7212
U(1)_Y coupling (GUT norm)		0.451900000000	0.451823783250	-168.6584
SU(2) fine-structure		0.033940000000	0.033914389437	-754.5835
U(1)_Y fine-structure (EW)		0.009745000000	0.009747192919	+225.0301
U(1)_Y fine-structure (GUT)		0.016241000000	0.016245321531	+266.0878
EM fine-structure (tree)		0.007575408000	0.007571189103	-556.9201
LEPTON YUKAWA COUPLINGS				
electron Yukawa		0.000002935028	0.000002935028	+0.0001
muon Yukawa		0.000606870766	0.000606870767	+0.0012
tau Yukawa		0.010205763474	0.010205761974	-0.1469
QUARK YUKAWA COUPLINGS				
up Yukawa		0.000012406407	0.000012476107	+5618.0642
down Yukawa		0.000026823112	0.000026802852	-755.3173
strange Yukawa		0.000536462247	0.000535616364	-1576.7799
charm Yukawa		0.007311739193	0.007319192576	+1019.3721
bottom Yukawa		0.024025926978	0.024002518423	-974.3039
top Yukawa		0.991191541612	0.991190871251	-0.6763
HIGGS SELF-COUPLING				
Higgs self-coupling		0.129280565411	0.129280113847	-3.4929
STRONG COUPLING				
strong coupling		0.118000000000	0.118047830923	+405.3468
PMNS NEUTRINO MIXING				
PMNS solar mixing fraction		0.303000000000	0.306578869942	+11811.4520
PMNS atmospheric mixing fraction		0.572000000000	0.571428571429	-999.0010
PMNS reactor mixing fraction		0.022030000000	0.021978021978	-2359.4200
PMNS CP phase	rad	3.4382986264	3.4557519189	+5076.1421
CKM QUARK MIXING				
Cabibbo mixing fraction		0.050640000000	0.050359712230	-5534.9086
CKM atmospheric mixing fraction		0.001682000000	0.001681796794	-120.8121
CKM reactor mixing fraction		0.000014600000	0.000014526054	-5064.8028
CKM CP phase	rad	1.1431906601	1.1519173063	+7633.5878
QCD THETA ANGLE				
QCD theta angle	rad	0.0000000000	0.0000000000	exact 0

XV. OPEN ITEMS AND FURTHER-EXPLORATION PROGRAM

We make this section explicit and visible. The numerical demonstration of the present paper is internally complete in the sense that, given the starting assumptions of Table I, no fitted continuous parameter is introduced and every reported value is a direct structural computation. The items below are open methodological questions that affect either the empirical CY3 identification of Sec. II B or the substrate-physical support of the starting assumptions themselves.

A. Pipeline state and in-flight remediation

We disclose the present state of the CY3 discrimination pipeline explicitly, separating closed results from indicative-but-unconverged results from in-flight work. The full engineering plan to upgrade the indicative results to converged-and-cross-checked status is the project's `REMEDIATION_PLAN_OPTION_B.md`, which specifies a seven-phase program with 111 positive and negative acceptance tests organised into five CI tiers.

a. Closed results (do not depend on in-flight work).

- *Catalogue uniqueness sweep at Hodge* $(3,3)$, mod-

TABLE XIV. Complete table (Part 2 of 2): substrate-framework predictions continued — massless gauge bosons, Newton's G, the Planck mass, anomalous magnetic moments, the Fermi constant, lepton lifetimes, W/Z/Higgs decay widths, Compton wavelengths and atomic constants, and the effective neutrino species count. Continuation of Table XIII.

Observable	Unit	Observed	Predicted	Deviation (ppm)
MASSLESS GAUGE BOSONS				
photon	MeV	0.000000	0.000000	exact 0
gluon 1	MeV	0.000000	0.000000	exact 0
gluon 2	MeV	0.000000	0.000000	exact 0
gluon 3	MeV	0.000000	0.000000	exact 0
gluon 4	MeV	0.000000	0.000000	exact 0
gluon 5	MeV	0.000000	0.000000	exact 0
gluon 6	MeV	0.000000	0.000000	exact 0
gluon 7	MeV	0.000000	0.000000	exact 0
gluon 8	MeV	0.000000	0.000000	exact 0
GRAVITATIONAL SECTOR				
Newton's G	m^3/kg/s^2	6.6743000000e-11	6.6743169083e-11	+2.5333
Planck mass	GeV	1.2208900000e22	1.2208885821e22	-1.1613
ANOMALOUS MAGNETIC MOMENTS				
electron anomalous moment		1.159652e-03	1.160667e-03	+875.2625
muon anomalous moment		1.165921e-03	1.166944e-03	+877.9375
tau anomalous moment		1.177210e-03	1.176417e-03	-673.9559
FERMI CONSTANT				
Fermi constant	GeV^-2	1.1663787000e-05	1.1663787073e-05	+0.0063
LEPTON LIFETIMES				
muon lifetime	s	2.196981e-06	2.197001e-06	+8.9421
tau lifetime	s	2.903000e-13	2.975067e-13	+24824.8892
W AND Z DECAY WIDTHS				
W total decay width	GeV	2.085000e+00	2.097035e+00	+5772.2721
Z total decay width	GeV	2.495500e+00	2.508963e+00	+5394.8742
Z invisible width (3 neutrinos)	GeV	4.995000e-01	4.976877e-01	-3628.3245
Z leptonic width (per lepton)	GeV	8.398000e-02	8.389803e-02	-976.0864
HIGGS DECAY WIDTHS				
Higgs width to bb	GeV	2.200000e-03	2.303033e-03	+46833.3164
Higgs width to tau tau	GeV	2.300000e-04	2.591192e-04	+126605.0368
Higgs width to cc	GeV	1.200000e-04	1.140530e-04	-49558.6271
Higgs total fermionic width	GeV	2.510000e-03	2.676205e-03	+66217.2986
COMPTON WAVELENGTHS AND ATOMIC CONSTANTS				
electron Compton wavelength	fm	2426.310239	2426.310132	-0.0440
muon Compton wavelength	fm	11.734441	11.734441	-0.0449
tau Compton wavelength	fm	0.697700	0.697771	+102.3542
Bohr radius	fm	52917.721090	52871.372618	-875.8592
Rydberg energy	eV	13.605693	13.629557	+1753.9345
EFFECTIVE NEUTRINO SPECIES				
effective neutrino species		3.046000000000	3.000000000000	-15101.7728

TABLE XV. Per-scheme within-scheme prediction table for the pole/direct scheme. Anchored on the Higgs vev; predictions for the remaining members computed as $m_X = (R_X/R_{\text{anchor}}) \cdot m_{\text{anchor}}$. $\Delta(\sigma_{\text{comb}})$ folds target and propagated-anchor uncertainty in quadrature. * marks the row used as the anchor for this scheme. Note that v_{Higgs} is not a direct experimental observable: the value 246.21965 GeV is back-computed structurally from the measured Fermi constant G_F via $v = (\sqrt{2}\,G_F)^{-1/2}$, so anchoring on the vev anchors on a structurally derived quantity rather than a raw measurement. Anchor: Higgs vev (observed = 246.2196500000 GeV, relative σ = 24.4 ppm).

Particle	Observed	σ_{obs} (ppm)	Predicted	Δ(pred−obs)	Δ (ppm)	Δ (σ_{comb})
electron	0.5109989500 MeV	+0.00	0.5109989500 MeV	0.0000000000 MeV	+0.0001	+0.0000
muon	105.6583755000 MeV	+0.00	105.6583756290 MeV	0.0000001290 MeV	+0.0012	+0.0001
tau	1776.8600000000 MeV	+67.53	1776.8597389792 MeV	-0.0002610208 MeV	-0.1469	-0.0020
top	172.5700000000 GeV	+1738.42	172.5698832877 GeV	-0.0001167123 GeV	-0.6763	-0.0004
W boson	80.3692000000 GeV	+161.75	80.3692003770 GeV	0.0000003770 GeV	+0.0047	+0.0000
Z boson	91.1876000000 GeV	+23.03	91.1900348152 GeV	0.0024348152 GeV	+26.7012	+0.7964
Higgs boson	125.2000000000 GeV	+878.59	125.1997813441 GeV	-0.0002186559 GeV	-1.7465	-0.0020
Higgs vev*	246.2196500000 GeV	+24.37	246.2196500000 GeV	–	–	–

ulo the gCICY non-manifest CAS residue. The Kreuzer–Skarke direct query returns `#NF: 0` at output limits up to $L = 10{,}000$; the $\chi = 0$ self-mirror diagonal sweep $h \in \{1, \ldots, 14\}$ confirms the bin is empty for $h \leq 13$ and populated from $h = 14$. The CICY, Schoen-class, Constantin–Lukas, Borisov–Caldararu Pfaffian, and gCICY-manifest classifications are programmatically checked.

- *Tian–Yau Yukawa-channel structural exclusion* at the searched bidegree range in the bicubic ambient at $h^{1,1} = 2$. No rank-3 SU(3) stable polystable monad bundle on TY/$\mathbb{Z}_3$ in the searched range sat-

TABLE XVI. Per-scheme within-scheme prediction table for the $\overline{\mathrm{MS}}$ scheme at $\mu = 2$ GeV (light quarks, single common scale – fully within-scheme). * marks the row used as the anchor for this scheme. Anchor: down (observed = 4.6700000000 MeV, relative $\sigma = 7.07e+04$ ppm).

Particle	Observed	σ_{obs} (ppm)	Predicted	Δ(pred−obs)	Δ (ppm)	Δ (σ_{comb})
up	2.1600000000 MeV	+175925.93	2.1737769100 MeV	0.0137769100 MeV	+6378.1991	+0.0336
down*	4.6700000000 MeV	+70663.81	4.6700000000 MeV	–	–	–
strange	93.4000000000 MeV	+92077.09	93.3232174033 MeV	-0.0767825967 MeV	-822.0835	-0.0071

TABLE XVII. Per-scheme within-scheme prediction table for the $\overline{\mathrm{MS}}$ scheme at $\mu = m_q$ (heavy quarks, per-particle scale – residual cross-scale within the group). * marks the row used as the anchor for this scheme. Anchor: bottom (observed = 4183.0000000000 MeV, relative $\sigma = 1.67e+03$ ppm).

Particle	Observed	σ_{obs} (ppm)	Predicted	Δ(pred−obs)	Δ (ppm)	Δ (σ_{comb})
charm	1273.0000000000 MeV	+3613.51	1275.5404247476 MeV	2.5404247476 MeV	+1995.6204	+0.5010
bottom*	4183.0000000000 MeV	+1673.44	4183.0000000000 MeV	–	–	–

isfies the heterotic-SM constraint suite; the leading rank-7 monad survivor $V_{\min 2}$ is excluded by $h^0(V_{\min 2}) = \mathbb{C}^2 \neq 0$.

- *E_8 Coxeter / McKay / closed-form algebraic identities* ($k_v = 1132/30$, $\delta_H = 1/59$, $\delta_W = 1/1121$, $\alpha_\tau = 1/278$, $\omega_{\mathrm{fix}} = 123/248$, $\alpha_s = 849/7192$, the gateway formula $F_e = (\dim -2)/\dim \cdot 5/(\dim \cdot h^M)/(1-1/(4(\dim -1)))$, etc.), all closed in rational arithmetic.

- *Mass-spectrum prediction pipeline* (`substrate_mass_spectrum.py`) producing the residuals reported in Tabs. XIII–XIV.

b. Indicative, not converged.

- *Chain-channel Bayes factor* $\ln \mathrm{BF} = -14.76$. The current chain-channel discriminators score the metric-Laplacian Galerkin residual against the framework's sub-Coxeter chain at $k_{\mathrm{test}} \in \{2, 3, 4, 5\}$. Both `p7_11_quark_chain.json` and `p7_11_lepton_chain.json` report `converged: false` with consecutive-degree relative deltas of order 11–35% on the Schoen-side residual. The figure is reported as supporting evidence; the converged replacement is Phase 2 of the remediation plan.

- *σ-channel matched-basis defense.* The current basis-matching at $n_b = 27$ is implemented as a lex-sorted truncation of TY's 87-dimensional invariant basis to Schoen's 27-dimensional basis. A geometric basis projection (eigenvalue-ranked principal components of the Laplacian, or harmonic-form ranking) is the in-flight Phase 3.3 upgrade; the σ-channel exclusion from the model-comparison BF is conditional on its survival under the upgrade.

- *σ functional definition.* The production code computes a weighted L^2 variance per Donaldson 2009 §3.4; a uniform pin of the L^1 vs L^2 functional choice across code, paper, and methodology table is in-flight Phase 3.1.

c. In-flight work (Option B remediation). The phases below are scheduled per the remediation plan. Each phase has explicit acceptance criteria gating the next.

1. *Phase 1 — pipeline integrity*: anomaly-cancellation regression test for the Schoen monad with $W = (27, 27)$; slope-stability proof on a stated Kähler-cone region; structure-group-of-V consistency (SU(3) vs SU(4) vs SU(5)); bundle-provenance forward-confirmation; replacement or ensemble-invariance of the LCG-random monad coefficients in `zero_modes.rs`; explicit Wilson-line representation specification.

2. *Phase 2 — real chain-channel Yukawa likelihood*: wire the existing `yukawa_pipeline.rs` (publication-grade end-to-end Yukawa pipeline already implemented in the rust_solver tree but not currently plumbed into the BF combiner) into `bayes_factor_multichannel.rs::from_yukawa_pipeline_pr` parallel TY implementation; bootstrap with ≥ 100 seeds; convergence guarantees with `Result<..., NotConvergedError>` short-circuit; Higgs-vev fitting/normalisation; two-loop SM RG running with threshold corrections.

3. *Phase 3 — σ-channel functional pin and matched-basis fix*: pin L^1 vs L^2; replace lex truncation with geometric basis projection; Donaldson convention robustness sweep across ≥ 3 published variants.

4. *Phase 4 — dark-matter $\Omega_{\mathrm{DM}} h^2$ re-derivation or relabelling*: the current pipeline uses the observed Planck 0.12 as a multiplicative prefactor; either re-derive the prefactor structurally or honestly relabel as `Omega_DM_h2_consistency_estimate` with the Planck empirical anchor disclosed.

TABLE XVIII. Supplementary derived neutrino-sector and cosmological observables not covered by the auto-generated complete table. Mass-squared splittings, sum of neutrino masses (Σm_ν), kinematic effective neutrino mass (m_β), neutrinoless double-beta effective mass ($m_{\beta\beta}$), and the dark-matter relic-density estimate $\Omega_{\rm DM}h^2$. e-anchor values are computed from the predicted neutrino masses; μ-anchor values are obtained by linear (m) or quadratic (m^2) rescaling under the new $h_{\rm eff}$ ratio.

Observable	Observed	e-anchor	μ-anchor	Deviation
Δm^2_{21} (eV^2)	$7.41^{+0.21}_{-0.20} \times 10^{-5}$	$7.4193221245 \times 10^{-5}$	$7.4193221075 \times 10^{-5}$	-90 ppm
Δm^2_{31} (eV^2)	$(2.511 \pm 0.027) \times 10^{-3}$	$2.5211270648 \times 10^{-3}$	$2.5211270590 \times 10^{-3}$	$+2440$ ppm
Σm_ν (meV)	< 72 [36]	59.2898456922	59.2898456244	within
m_β (meV)	< 450 [37]	8.8276108075	8.8276107974	within
$m_{\beta\beta}$ (meV)	< 28–122 [38]	3.9984876767	3.9984876721	within
$\Omega_{\rm DM}h^2$	0.1200 ± 0.001 [39]	0.119981783712	0.119981783369	-151.8 ppm

5. *Phase 5 — structural-rule generators* for the per-particle Coxeter chain indices k_X, the per-quark sign assignments χ_q, λ_q, and the rational closures used in $\sin^2\theta_W$ and $\alpha_s(M_Z)$; or honest documentation as encoded structural-rule-consistent choices in a data file.

6. *Phase 6 — convergence, reproducibility, external verification*: programmatic gCICY $(3,3)$ cycle 2.6 in a Sage/Macaulay2-equipped container; adjacent-Hodge-bin sweep at $(3,4),(4,3)$; rational-search baseline for closure rules; Zenodo-archived reproducibility kit (`cargo run --release --bin reproduce_paper_results`); external independent re-implementation in the AKLP PyTorch suite or equivalent; pre-registration of falsifiable predictions.

7. *Phase 7 — paper/pipeline alignment audit*: pre-submission line-by-line audit against a 111-test acceptance matrix.

The acceptance criteria, decision points, risk register, and test matrix for each phase are detailed in `REMEDIATION_PLAN_OPTION_B.md`. Estimated total effort: 6–12 person-weeks; submission-readiness gate is Phase 7 sign-off.

B. Open methodological items

1. *gCICY non-manifest residue at Hodge* $(3,3)$. The catalogue uniqueness sweep (Sec. II B) closes the polytope catalogue [19] and the published gCICY classifications [20–22] at $(h^{1,1}, h^{2,1}) = (3,3)$, but a future codim $\geq (3,1)$ extension producing a $(3,3)$ entry is not pre-empted. *Required work:* structural enumeration over the gCICY-extension constructions at parity with the polytope sweep, applying the same four-principle filter and reporting non-Schoen survivors if any. Until this is done the catalogue uniqueness statement at $(3,3)$ is conditional on the gCICY non-manifest residue.

2. *Donaldson convention robustness on the σ-channel discrimination.* The $|t| = 6.92$ Schoen-vs-TY σ-channel result reported in Sec. II B depends on the specific choice of Donaldson h-inversion procedure and the Kähler-form projection convention used in the metric pipeline. The paired-bootstrap sign-reversal defense addresses the basis-size artifact at matched basis size; it does not address convention-choice artifacts. *Required work:* robustness sweep over ≥ 3 published convention variants for both the h-inversion handling and the Kähler-form projection, with a stress-test report on whether the $|t| = 6.92$ result survives or degrades. The σ-channel discrimination should be re-stated with an explicit convention sensitivity envelope.

3. *Schoen-uniqueness sweep at adjacent Hodge bins.* The catalogue uniqueness statement is at the single Hodge bin $(3,3)$. *Required work:* extension of the four-cycle sweep to bins adjacent to $(3,3)$ (in particular $(3,4)$ and $(4,3)$) at the same fidelity, to confirm that the empty-bin result at $(3,3)$ is not isolated to one bin under the four-principle filter.

4. *Bundle provenance forward-confirmation.* An audit performed during the development of this paper retracted some previously asserted bundle provenance to nominally cited references; bundles currently in use have been re-pointed to primary sources. *Required work:* a comprehensive forward-confirmation that, for every bundle data set used in the present paper, the citation chain terminates at a primary source where the bundle is constructed and its stability/topological constraints are stated.

C. Open framework items deferred to companion paper

The following items are starting commitments of the present paper and are listed here for transparency. Their substrate-physical support — i.e. the argument that one would expect these and not other substrate-physical structures — is the subject of the companion paper [43] and of the foundational monograph [42], and is not load-bearing for the numerical demonstration of this paper.

1. *Heterotic $E_8 \times E_8$ doubling.* The Coldea *et al.* [3] $CoNb_2O_6$ measurement empirically anchors one E_8 factor. The doubling to $E_8 \times E_8$ is a heterotic-string commitment [11] adopted here; an independent empirical anchor for the second E_8 factor is not claimed.

2. *McKay-icosahedral seed.* The selection of the binary icosahedral McKay-dual of E_8 as the framework's harmonic seed, and the resulting φ-chain / $\sqrt{2}$-chain decomposition, is a structural commitment.

3. *Gateway-formula structural integers.* The factor of 4 in the sub-harmonic loss 988/987, the gateway-depth $M = 14$, and the boundary capacity $h + m_{\max} = 59$ used in the dark-sector coupling family admit dual-sector / two-orientation readings within the framework but are not derived in the present paper from a specific group-theoretic counting rule on the Schoen quotient.

4. *Equivariant-cohomology identity for $\theta_{\rm QCD} = 0$.* The vanishing of the strong-CP angle is presented in Sec. VII contingent on a $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient-projected twisted-trace cancellation. The identity itself is adopted as input from the framework; an explicit proof on the Schoen $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$ quotient is part of the open program.

5. *The $\omega_{\rm fix} = 123/248$ identity is algebraic, not eigenvalue.* $\omega_{\rm fix} = (\dim -2)/(2{\cdot}\dim)$ appears in the gateway formula as an exact E_8-algebraic coefficient. It is *not* an eigenvalue of any operator and is not predicted to be empirically measurable as such. We have internally tested and retracted the alternative hypothesis under which it would be an eigenvalue; the algebraic reading is the correct one and is the only role $\omega_{\rm fix}$ plays in the construction.

6. *Hodge-pair self-conjugacy $h^{1,1} = h^{2,1}$.* The substrate-physical argument that the framework selects self-conjugate Hodge structures (and hence $(3,3)$ as the mathematical-minimum admissible bin compatible with three generations and Wilson-line breaking) lives in the companion paper [43]. The present paper takes the $(3,3)$ outcome as the input to Sec. II B.

D. What this paper does not claim

For the avoidance of overreach we state plainly:

- This paper does not claim that the starting assumptions of Table I are forced from first principles in the sense of being derivable from inside the framework's macroscopic 4D regime. From a 4D observer's vantage they are evaluated *empirically* — by the predictions they generate.
- The 26-to-zero-free-parameter reduction reported here is *conditional* on the starting assumptions; without those assumptions the present construction makes no statement.
- Schoen uniqueness in full generality across the Calabi–Yau threefold landscape is not claimed; the catalogue uniqueness statement is at the searched Hodge bin and is conditional on the gCICY residue (item 1 of Sec. XV B).
- Internal consistency of the present construction at the 0.0298 ppm ($\sim$ 30 ppb) two-anchor level (Sec. XII) is a self-consistency statement of the structural pipeline; it is not itself a derivation of the starting assumptions.
- No claim is made that the substrate-physical readings of c, $\hbar$, and the macroscopic emergence of GR/QM/Newton/Maxwell from substrate elasticity, deferred to the companion paper [43], are required for the numerical demonstration of this paper. They are interpretive context; the numerical results stand or fall on their own.

E. Further-exploration program

The most informative empirical and theoretical extensions of the present construction, in priority order:

1. Direct dark-sector searches at the predicted dominant order-5 slot $m_5 = 602.645$ neV with $g_{a\gamma\gamma} = 1.78 \times 10^{-17}$ GeV^{-1}. A null result at the predicted mass and coupling, with sensitivity exceeding the prediction, falsifies the dark-sector construction.

2. Improved precision measurements on m_W, m_H, m_t, $\sin^2\theta_W$, m_β (KATRIN), and Σm_ν (cosmological) at thresholds that either tighten the present ppm/ppb-level residuals or break them.

3. Independent reproduction of the Schoen-vs-TY σ-channel discrimination pipeline by a third party, using independent metric codes and convention sweeps.

4. Closure of the gCICY non-manifest residue (Sec. XV B item 1) and the adjacent-Hodge-bin sweep (item 3).

5. Heterotic-anomaly index or $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient-trace derivation of the gateway-formula structural integers (Sec. XV C item 3).

6. Explicit Schoen-side proof of the $\mathbb{Z}_3 \times \mathbb{Z}_3$ quotient-projected twisted-trace cancellation underlying $\theta_{\rm QCD} = 0$ (Sec. XV C item 4).

7. Peer-reviewed development of the foundational arguments relegated to the companion paper [43].

XVI. DISCUSSION

A. Parameter-count semantics

Conditional on the starting assumptions of Table I, the construction presented here contains *no fitted continuous parameter*: the 26 continuous parameters of the SM are computed structurally from the algebraic data of the starting assumptions, and the residuals reported in Sec. XIII are predictions versus measurement, not fits versus measurement. We are explicit about the scope of this statement. (i) Without the starting commitments (Table I) the construction has no content; the reduction is not a model-independent claim, but a zero-free-parameter consequence of those commitments. (ii) Two sectors carry residual conditionality even within the framework: the dark-matter abundance closes through a substrate-to-misalignment dictionary whose substrate-physical derivation lives in the companion paper, and the strong-CP angle closes through the quotient-projected twisted-trace cancellation described in Sec. VII and listed in Sec. XV C. (iii) The argument that the starting commitments themselves are minimal and substrate-physically expected is the subject of the companion paper [43] and is not load-bearing here.

B. Common geometric origin

Notable unifying features include both CP phases as rational multiples of π ($\delta_{\rm CP} = \pi + \pi/10$, $\delta_{\rm CKM} = 11\pi/30$), the dark-sector coupling family graded by E_8 exponents ($\delta_H = 1/59$, $\delta_W = 1/(59 \cdot 19)$), the quotient-based conditional route to a vanishing strong CP angle, the unified neutrino-dark-matter tower, and the strong coupling as the rational 849/7192 from forced E_8 integers.

C. Comparison to prior work

The closest empirical precedent is Coldea *et al.* [3]'s confirmation of the E_8 mass spectrum in $CoNb_2O_6$. The closest theoretical precedent is Koide's mass formula [41], which relates the three charged-lepton masses at 10^{-5} precision with no derivation. The present framework matches Koide-level precision on leptons and extends the structural construction to all 26 SM parameters plus dark matter and gravity, while leaving the compactification-level explanation of some closure factors for follow-up work.

The geometric strong-CP mechanism, if the quotient-projected twisted trace vanishes, avoids invoking Peccei–Quinn [30] and so contrasts with the dominant axion-mediated approach.

D. Open directions

The complete enumeration of open methodological items, deferred framework commitments, and the further-exploration program is given in Sec. XV. Among the items most likely to sharpen or break the present construction:

1. Direct dark-sector searches at the predicted dominant order-5 slot (Sec. XV E item 1).

2. Closure of the gCICY non-manifest residue and the adjacent-Hodge-bin sweep (Sec. XV B items 1 and 3).

3. Donaldson-convention robustness sweep on the σ-channel discrimination (Sec. XV B item 2).

4. Substrate-native cosmological evolution for the dark-matter tower, replacing the misalignment-style abundance estimate with a full continuum calculation.

5. Cosmological inflationary connection: $n_s = 58/60$ from dual-E_8 Coxeter structure.

E. Conclusion

Conditional on the starting assumptions of Table I — heterotic $E_8 \times E_8$ on the Schoen Calabi–Yau quotient $X/(\mathbb{Z}_3 \times \mathbb{Z}_3)$, the McKay-icosahedral seed, the phenomenological $\varphi/\sqrt{2}$ chain split for the lepton and quark/EW-boson sectors, the E_8 Coxeter spectrum, and the substrate-tension reading of the Higgs vacuum — the present paper computes the SM mass spectrum, mixing parameters, gauge couplings, the strong-CP angle, the cosmological dark-matter abundance, and Newton's G *directly*, with no fitted continuous parameter and no per-particle scheme choice. The Schoen-versus-Tian–Yau σ-channel discrimination (Sec. II B) and the four-cycle catalogue uniqueness sweep at Hodge $(3, 3)$ provide the empirical CY3 identification, with explicit hedges on the gCICY residue and the Donaldson convention sensitivity (Sec. XV). The numerical residuals against PDG and Planck 2018 measurements are predictions, not fits. From a 4D observer's vantage the starting assumptions cannot be *proven*; this paper evaluates them by empirical consistency, and reports the result. The companion paper [43] addresses the separate question of whether those starting assumptions are minimal and substrate-physically expected; that argument is independent of the present demonstration and is not relied upon here.

TABLE XIX. Coxeter step assignments. The φ-chain hosts integer E_8 exponent positions; the $\sqrt{2}$-chain hosts integer and half-integer positions. Cycle 2 indicates positions beyond $h = 30$.

Particle	k	Chain type / sector
Electron	0	φ gateway, dark-sector
Muon	11	φ exponent, visible
Tau	17	φ exponent, visible
Up	4	$\sqrt{2}$ int
Down	6.5	$\sqrt{2}$ half
Strange	15	$\sqrt{2}$ midpoint
Charm	22.5	$\sqrt{2}$ half
Bottom	26	$\sqrt{2}$ int
Top	36.5	$\sqrt{2}$ half, cycle 2
W boson	34.5	cycle 2, with δ_W
Z boson	via $m_W/\cos\theta_W$	derived
Higgs vev	1132/30	cycle 2
Higgs boson	$k_v - 2$	breathing mode, with δ_H

Appendix A: Coxeter step assignments

Appendix B: Complete structural correction list

Appendix C: Numerical precision

All numerical evaluations use mpmath at $\geq$ 50-digit decimal precision. Key exact rational values:

- $h_{\text{eff}}^{(e)}/m_e = 124/123$;
- $30^{14} = 4{,}782{,}969 \times 10^{14}$;
- $\sin^2\theta_{12}^{\text{CKM}} = 14/278 = 7/139$;
- $\delta_{\text{CKM}} = 11\pi/30$;
- $\delta_{\text{CP}} = 11\pi/10$;
- $\sin^2\theta_W = 2/9 + 1/[h(h+\varphi^2)]$;
- $\alpha_s(M_Z) = 849/7192$;
- $\delta_H = 1/59$, $\delta_W = 1/1121$.

[1] S. Navas *et al.* (Particle Data Group), "Review of Particle Physics," Phys. Rev. D **110**, 030001 (2024). Open-access PDG portal (free, complete review): pdg.lbl.gov/2024. DOI: 10.1103/PhysRevD.110.030001.

[2] G. Bertone and T. M. P. Tait, "A new era in the search for dark matter," Nature **562**, 51–56 (2018). Open-access preprint (free): arXiv:1810.01668. DOI: 10.1038/s41586-018-0542-z.

[3] R. Coldea, D. A. Tennant, E. M. Wheeler, E. Wawrzynska, D. Prabhakaran, M. Telling, K. Habicht, P. Smeibidl, and K. Kiefer, "Quantum Criticality in an Ising Chain: Experimental Evidence for Emergent E_8 Symmetry," Science **327**, 177–180 (2010). Open-access repository copy (free, same content): arXiv:1103.3694. PubMed (free abstract): PMID 20056884. DOI: 10.1126/science.1180085.

[4] A. B. Zamolodchikov, "Integrals of motion and S-matrix of the (scaled) $T = T_c$ Ising model with magnetic field," Int. J. Mod. Phys. A **4**, 4235–4248 (1989). Inspire-HEP record (free abstract, cross-references): inspirehep.net/literature/276727. DOI: 10.1142/S0217751X8900176X.

[5] J. McKay, "Graphs, singularities, and finite groups," Proc. Symp. Pure Math. **37**, 183–186 (Amer. Math. Soc., 1980). Open scan of the original conference paper (free): McKaysOriginalPaper.pdf. DOI of the original: 10.1090/pspum/037/604577.

[6] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics **29** (Cambridge University Press, 1990). ISBN 978-0-521-43613-7. Identity $|\Phi| = h \cdot \text{rank}$ for irreducible finite Coxeter groups: §3.18, Eq. (3) (consequence of the exponents satisfying $\sum_j (e_j + 1) = |\Phi|/2 + \text{rank}$ with mean exponent $h/2$). Cambridge product page (free abstract): cambridge.org/core/books/reflection-groups-and-coxeter-groups. DOI: 10.1017/CBO9780511623646.

[7] P. W. Anderson, "Plasmons, Gauge Invariance, and Mass," Phys. Rev. **130**, 439–442 (1963). APS abstract page (free abstract): journals.aps.org/pr/abstract/10.1103/PhysRev.130.439. NASA ADS (free abstract + cross-references): ui.adsabs.harvard.edu/abs/1963PhRv..130..439A. DOI: 10.1103/PhysRev.130.439.

[8] G. E. Volovik, *The Universe in a Helium Droplet*, International Series of Monographs on Physics **117** (Oxford University Press, 2003). Oxford Academic landing page (free metadata): academic.oup.com/book/11557. DOI of book: 10.1093/acprof:oso/9780199564842.001.0001.

[9] G. E. Volovik, "Superfluid analogies of cosmological phenomena," Phys. Rep. **351**, 195–348 (2001). Open-access preprint (free): arXiv:gr-qc/0005091. DOI: 10.1016/S0370-1573(00)00139-3.

[10] C. Schoen, "On fiber products of rational elliptic surfaces with section," Math. Z. **197**, 177–199 (1988). EuDML page (free metadata): eudml.org/doc/183725. DOI: 10.1007/BF01215188.

[11] D. J. Gross, J. A. Harvey, E. Martinec, and R. Rohm, "Heterotic String," Phys. Rev. Lett. **54**, 502–505 (1985). Inspire-HEP record (free abstract + cross-references): inspirehep.net/literature/15974. NASA ADS: ui.adsabs.harvard.edu/abs/1985PhRvL..54..502G. DOI: 10.1103/PhysRevLett.54.502.

[12] V. Braun, Y.-H. He, B. A. Ovrut, and T. Pantev, "A standard model from the $E_8 \times E_8$ heterotic superstring," JHEP **06**, 039 (2005). Open-access preprint (free): arXiv:hep-th/0502155. DOI: 10.1088/1126-6708/2005/06/039.

[13] R. Donagi, Y.-H. He, B. A. Ovrut, and R. Reinbacher, "The spectra of heterotic standard model vacua," JHEP

TABLE XX. All structural corrections, built from framework-fixed E_8 invariants dim $= 248$, $h = 30$, $M = 14$, rank $= 8$, $m_{\max} = 29$, together with the framework-selected McKay order 5 and golden ratio φ.

Symbol	Definition	Role
Δ_φ	$1/(h+\varphi^2)$	enters $\sin^2\theta_W$
Δ_2	$1/2^5 = 1/32$	2-fold structure
α_τ	$1/(\dim + h) = 1/278$	tau base
$\alpha_\tau^{\text{full}}$	$1/(\dim + h + \varphi^2)$	tau closed-form
α_μ	$2\alpha_\tau^2$	muon
α_v	$(\operatorname{rank} h/m_{\max})\alpha_\tau^2/(1+\alpha_\tau/(2\varphi))$	Higgs vev (closure 2φ)
α_H	$((\operatorname{rank}-2)h/m_{\max})\alpha_\tau^2/(1+\alpha_\tau/h_{SU(3)})$	Higgs boson (closure $h_{SU(3)}$)
α_W	$(2/e_2)\alpha_\tau^2/(1+\alpha_\tau/h_{SU(3)})$	W boson sub-harmonic
α_t	$(h/h_{SU(3)})^2\alpha_\tau^2/(1+\alpha_\tau/h_{SU(3)})$	Top quark sub-harmonic
δ_H	$1/(h+m_{\max}) = 1/59$	Higgs dark coupling
δ_W	$1/(59\cdot 19) = 1/1121$	W dark coupling
ε	$1/(2\dim) = 1/496$	per-step quark loss
ω_{fix}	$1/2 - 1/\dim = 123/248$	gateway normalization
k_v	$1132/30$	Higgs vev position
f_5	$\frac{\operatorname{rank}(h-\operatorname{rank})}{\dim}\frac{\sqrt{vM_{\text{Pl}}}}{\alpha_\tau}$	DM decay constant
θ_5	$\arctan\varphi$	DM misalignment

06, 070 (2005). Open-access preprint (free): arXiv:hep-th/0411156. DOI: 10.1088/1126-6708/2005/06/070.

[14] V. Braun, Y.-H. He, B. A. Ovrut, and T. Pantev, "Vector Bundle Extensions, Sheaf Cohomology, and the Heterotic Standard Model," Adv. Theor. Math. Phys. **10**, 525–589 (2006). Open-access preprint (free): arXiv:hep-th/0505041. §6 introduces the rank-4 $SU(4)$ extension construction used as the framework's bundle commitment; Eq. 99 records the heterotic Bianchi anomaly-cancellation identity with the positive-tension 5-brane wrapping τ_1^2.

[15] J. C. Pati and A. Salam, "Lepton number as the fourth color," Phys. Rev. D **10**, 275–289 (1974); erratum *ibid.* **11**, 703 (1975). APS abstract page (free abstract): journals.aps.org/prd/abstract/10.1103/PhysRevD.10.275. DOI: 10.1103/PhysRevD.10.275. Source for the substrate-physical "lepton number as the fourth colour" reading of the rank-4 $SU(4)$ internal-symmetry structure committed in this paper.

[16] G. Tian and S.-T. Yau, "Three-dimensional algebraic manifolds with $c_1 = 0$ and $\chi = -6$," in *Mathematical Aspects of String Theory* (San Diego, CA, 1986), Adv. Ser. Math. Phys. Vol. 1, World Scientific, Singapore (1987).

[17] L. B. Anderson, J. Gray, A. Lukas, and E. Palti, "Heterotic line bundle standard models," JHEP **06**, 113 (2012). Open-access preprint (free): arXiv:1202.1757. DOI: 10.1007/JHEP06(2012)113.

[18] L. B. Anderson, Y.-H. He, and A. Lukas, "Heterotic compactification, an algorithmic approach," JHEP **07**, 049 (2007). Open-access preprint (free): arXiv:hep-th/0702210. DOI: 10.1088/1126-6708/2007/07/049.

[19] M. Kreuzer and H. Skarke, "Complete classification of reflexive polyhedra in four dimensions," Adv. Theor. Math. Phys. **4**, 1209–1230 (2000). Open-access preprint (free): arXiv:hep-th/0002240. Catalogue front-end: `http://hep.itp.tuwien.ac.at/~kreuzer/CY/`.

[20] L. B. Anderson, F. Apruzzi, X. Gao, J. Gray, and S.-J. Lee, "A new construction of Calabi–Yau manifolds: generalized CICYs," Nucl. Phys. B **906**, 441–496 (2016). Open-access preprint (free): arXiv:1507.03235. DOI: 10.1016/j.nuclphysb.2016.03.016.

[21] M. Larfors and R. Schneider, "Line bundle cohomologies on CICYs with Picard number two," Fortsch. Phys. **67**, 1900083 (2019). Open-access preprint (free): arXiv:2003.10130.

[22] A. Constantin, A. Lukas, and C. Mishra, "The family of quintic Calabi–Yau threefolds, free quotients, and their cohomology," Fortsch. Phys. **64**, 463–509 (2016). Open-access preprint (free): arXiv:1606.04032.

[23] D. Huybrechts and M. Lehn, *The Geometry of Moduli Spaces of Sheaves*, 2nd ed., Cambridge Mathematical Library, Cambridge University Press (2010). DOI: 10.1017/CBO9780511711985.

[24] R. E. Kass and A. E. Raftery, "Bayes factors," J. Amer. Statist. Assoc. **90**, 773–795 (1995). DOI: 10.1080/01621459.1995.10476572.

[25] S. K. Donaldson, *Some numerical results in complex differential geometry*, Pure Appl. Math. Q. **5** (2009), 571–618. DOI: 10.4310/PAMQ.2009.v5.n2.a2. arXiv:math/0512625. See §3 eq. (3.4) for the weighted L^2 Monge–Ampère residual functional $\sigma^2 = \langle(\eta/\kappa - 1)^2\rangle_w$ with $\kappa = \langle\eta\rangle_w$ that defines the production σ-discriminator used in this work.

[26] H. Jeffreys, *Theory of Probability*, 3rd ed. (Oxford University Press, 1961). Source of the "decisive evidence" threshold $\log_{10}\mathrm{BF} > 2$ on the Bayes-factor strength scale.

[27] S. K. Donaldson, "Anti self-dual Yang–Mills connections over complex algebraic surfaces and stable vector bundles," Proc. London Math. Soc. **50**, 1–26 (1985). DOI: 10.1112/plms/s3-50.1.1.

[28] K. Uhlenbeck and S.-T. Yau, "On the existence of Hermitian–Yang–Mills connections in stable vector bundles," Comm. Pure Appl. Math. **39**, S257–S293 (1986). DOI: 10.1002/cpa.3160390714.

[29] J. E. Kim and G. Carosi, "Axions and the strong CP problem," Rev. Mod. Phys. **82**, 557–601 (2010). Open-access preprint (free): arXiv:0807.3125. DOI: 10.1103/RevModPhys.82.557; Erratum: Rev. Mod. Phys. **91**, 049902 (2019).

[30] R. D. Peccei and H. R. Quinn, "CP Conservation in the Presence of Pseudoparticles," Phys. Rev. Lett. **38**, 1440–1443 (1977). NASA ADS (free abstract + cross-references): ui.adsabs.harvard.edu/abs/1977PhRvL..38.1440P. Inspire-HEP: inspirehep.net/literature/120612. DOI: 10.1103/PhysRevLett.38.1440.

[31] C. Abel *et al.* (nEDM Collaboration at PSI), "Measurement of the Permanent Electric Dipole Moment of the Neutron," Phys. Rev. Lett. **124**, 081803 (2020). Open-access preprint (free): arXiv:2001.11966. DOI: 10.1103/PhysRevLett.124.081803. Reports $|d_n| < 1.8 \times 10^{-26}\, e \cdot \mathrm{cm}$ at 90% C.L.

[32] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, "CODATA recommended values of the fundamental physical constants: 2022," Rev. Mod. Phys. **97**, 025002 (2025). Open-access preprint (free): arXiv:2409.03787. NIST free interactive constant database: physics.nist.gov/cuu/Constants. DOI: 10.1103/RevModPhys.97.025002.

[33] C. Rothleitner and S. Schlamminger, "Invited Review Article: Measurements of the Newtonian constant of gravitation, G," Rev. Sci. Instrum. **88**, 111101 (2017). NIST author manuscript (free): nist.gov/publications/invited-review-article-measurements-newtonian-constant-gravitation-g. DOI: 10.1063/1.4994619.

[34] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou, "The fate of hints: updated global analysis of three-flavor neutrino oscillations," JHEP **09**, 178 (2020). JHEP is open access; published version free at: doi:10.1007/JHEP09(2020)178. Preprint: arXiv:2007.14792.

[35] NuFIT Collaboration (I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, A. Zhou), "NuFIT 5.3: Three-neutrino fit based on data available in March 2024," `http://www.nu-fit.org/?q=node/278` (free, accessed April 2026).

[36] A. G. Adame *et al.* (DESI Collaboration), "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations," JCAP **02**, 021 (2025). JCAP is open access; published version free at: doi:10.1088/1475-7516/2025/02/021. Preprint: arXiv:2404.03002.

[37] M. Aker *et al.* (KATRIN Collaboration), "Direct neutrino-mass measurement based on 259 days of KATRIN data," Science **388**, 180–185 (2025). Open-access preprint (free): arXiv:2406.13516. DOI: 10.1126/science.adq9592. Reports $m_\beta < 0.45$ eV at 90% C.L.

[38] S. Abe *et al.* (KamLAND-Zen Collaboration), "Search for Majorana Neutrinos with the Complete KamLAND-Zen Dataset," Open-access preprint (free): arXiv:2406.11438 (2024). Reports $T^{0\nu}_{1/2} > 3.8 \times 10^{26}$ yr at 90% C.L., $\langle m_{\beta\beta} \rangle < 28$–$122$ meV depending on nuclear matrix element.

[39] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020). A&A is open access; published version free at: aanda.org/articles/aa/full_html/2020/09/aa33910-18. Preprint: arXiv:1807.06209. DOI: 10.1051/0004-6361/201833910; Erratum: A&A **652**, C4 (2021).

[40] K. K. Rogers and H. V. Peiris, "Strong Bound on Canonical Ultralight Axion Dark Matter from the Lyman-Alpha Forest," Phys. Rev. Lett. **126**, 071302 (2021). Open-access preprint (free): arXiv:2007.12705. DOI: 10.1103/PhysRevLett.126.071302. Reports $m_a > 2 \times 10^{-20}$ eV at 95% C.L.

[41] Y. Koide, "Fermion-boson two-body model of quarks and leptons and Cabibbo mixing," Lett. Nuovo Cimento **34**, 201–205 (1982). Inspire-HEP record (free abstract): inspirehep.net/literature/176414. DOI: 10.1007/BF02817096; expanded in Y. Koide, Phys. Rev. D **28**, 252 (1983), APS abstract; DOI: 10.1103/PhysRevD.28.252; Phys. Lett. B **120**, 161–165 (1983), publisher abstract. DOI: 10.1016/0370-2693(83)90644-5.

[42] E. Gable, *Exist: The Universal Ontology* (2026). Foundational substrate-physical derivations of c, $\hbar$, general relativity, quantum mechanics, Newton's laws, Maxwell's equations, and the heterotic $E_8 \times E_8$ commitment.

[43] E. Gable, "The unreasonable effectiveness of mathematics, solved: substrate-physical derivation of the first-principles starting assumptions," companion paper (in preparation, 2026). Argues that the starting assumptions of Table I are minimal and substrate-physically expected. Independent of, and not load-bearing for, the numerical demonstration of the present paper.

[44] V. I. Arnold, *Catastrophe Theory*, 3rd ed. (Springer-Verlag, Berlin, 1992). Free expository synonym (open access): encyclopediaofmath.org/wiki/Catastrophe_theory (Springer/EMS, free). Springer landing page (free metadata): link.springer.com/book/10.1007/978-3-642-58124-3. DOI of book: 10.1007/978-3-642-58124-3. The simply-laced ADE classification of elementary catastrophes shares its label set with the McKay correspondence used in Sec. II, but the two structures are mathematically distinct.

[45] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," Classical Quantum Gravity **11**, L73–L77 (1994). Open-access preprint (free): arXiv:gr-qc/0009013. DOI: 10.1088/0264-9381/11/5/001.

[46] P. Touboul *et al.* (MICROSCOPE Collaboration), "MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle," Phys. Rev. Lett. **129**, 121102 (2022). Open-access preprint (free): arXiv:2209.15487. DOI: 10.1103/PhysRevLett.129.121102. Reports Eötvös parameter $\eta(\mathrm{Ti}, \mathrm{Pt}) = (-1.5 \pm 2.3_{\mathrm{stat}} \pm 1.5_{\mathrm{sys}}) \times 10^{-15}$.